P1-1

Instinct Automated Surveying Unit (ASU) – Two seasons of optimizing PAM with the world's first real-time, satellite-enabled ARU

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Historically, wildlife surveying for Northern Spotted Owls (Strix occidentalis caurina), and other species of interest globally largely involved manual active surveying which is time/labor-intensive, error-prone and negatively impacts the target species. Autonomous Recording Units (ARUs) present the option of passive acoustic monitoring (PAM), but using the large amounts of audio data generated is a cumbersome process which introduces further error, and call detections are only known weeks, months or even a year after the initial recording was produced, limiting feasibility of passive surveying for research and forest management purposes. In April 2024, Instinct launched the Automated Surveying Unit (ASU) - the world's first satellite-enabled, edge-computing ARU for real-time, bioacoustic monitoring. The ASU records audio like ARUs, but then also processes and classifies sounds in the audio immediately after it is captured, and then transmits detection counts via satellite so species detections are known within hours of being recorded from any site, no matter how remote. Instinct deployed dozens of ASUs across the U.S west coast with customers for the 2024 survey season. A full season of operation yielded results and observations which highlighted potential improvement and expansion. Therefore, Instinct launched the ASU 2 in April 2025. Improvements across different modules of the device make it much more user-friendly, powerful, efficient and economical. The ASU addresses shortcomings associated with passive monitoring, and will make it easier for the wildlife biology community to carry out surveying and research, helping us better protect and promote endangered species on our planet.

P1-2

Active sampling for data efficient large-scale bioacoustic monitoring

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Passive acoustic monitoring can provide biodiversity insights at previously infeasible scales and resolutions, but also produce volumes of data which necessitate the use of automatic classification methods. Pre-trained neural network models provide generalisable performance across many species but lack reliability especially given challenges such as severe class imbalance, geographic and intra-species variability. Fine-tuning of application specific models can improve reliability but become difficult to scale due a lack of labeled data. Active learning is a technique which optimises model performance with minimal samples by tactically selecting the most informative samples for annotation. Active learning has not yet been applied to large-scale (transnational) bioacoustic monitoring of avian species. Here we show the efficacy of entropy-based active sampling of bioacoustic field data across broad spatial and temporal scales. Our results demonstrate a clear improvement in reliability compared to pre-trained model performance (BirdNet) and also provide insight into how active learning can improve the scalability and data efficiency of inference. We demonstrate that training is sensitive to selection of the first batch, suggesting the prioritisation of annotation resources at active learning cycle onset. We recommend weight initialisation and regularisation approaches, showing improved robustness and overall model performance. We also recommend initial selection using confidence sampling and diversification methods for cases where initialisation is not feasible. These findings support the use of active learning as a scalable, data-efficient approach for ecological monitoring, enabling reliable biodiversity assessments across broad spatial and temporal scales.

P1-3

Seasonal vocal activity of the endemic Xantus's hummingbird in a protected natural area using passive acoustic monitoring

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The information about the acoustic repertoire of the Xantus's hummingbird (Basilinna xantusii) is limited; however, confirmed recordings in repositories show distinctive trill calls composed of several pulses. This species is endemic to the central and southern Baja California peninsula in Mexico; little is known about its biology and ecology, and no bioacoustics research have been conducted to date. Therefore, the aim of this study was to describe and compare its vocal activity in two natural environments. Considering a year cycle, continuous 48-h passive acoustic recordings were taken bimonthly in two ecosystems (tropical deciduous forest and oak forest) located in Sierra La Laguna Biosphere Reserve. A total of 2,720 calls were registered, 75% of them in the tropical deciduous forest. Calling activity peaked in the morning between 6 and 10 a.m. (68%). In both environments, calls were registered year-round. More calls were detected during the warm season and fewer during the cold season in the oak forest (900 m altitude), whilst in the tropical deciduous forest (500 m altitude) most calls were recorded during the cold season. These results expand our understanding of the vocal activity of this endemic species present in this protected natural area, confirming it as resident, as well as highlight passive acoustic monitoring as a useful tool for studying hummingbird ecology.

P1-4

Derivation of acoustic indices related to biodiversity from wideband environmental sounds in tropical rain forests of Borneo

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Ultra-wideband recordings of environmental sounds were made for three to five days at one to three locations in three protected forest areas in the tropical rainforest, where the land environment and biodiversity are likely to differ. Eight types of acoustic indices related to biodiversity, Acoustic Complexity (AC), Acoustic Evenness (AE), Acoustic Diversity (AD), Frequency-dependent AD (FAD), acoustic entropy (H), spectral entropy (Hf), temporal entropy (Ht), and Bioacoustic (BA) indices were calculated from multiple frequency bands, including below 20 kHz and above, and their diurnal variations were focused on. Past studies have not derived acoustic indices above 20 kHz, although the tropical rainforest contains rich biological sounds above 20 kHz. Recordings were conducted by the AudioMoth recorders at a sampling frequency of 384 kHz. In the two protected forest areas, the three recorders were set about 100 m apart, and recordings were made for three or five days, while in the other area, only one recorder was set, and recordings were made for three days. The recorders recorded for 5 minutes every 30 minutes. The aim was to find acoustic indices and their frequency bands representing different sound environments by showing different diurnal variations between protected forest areas and similar diurnal variations between different recording days and locations within the same area. As a result, the diurnal variations of the maximum temporal entropy index calculated every minute for over 5 minutes from the 200 Hz to 96 kHz band mostly satisfied the above criteria and could express differences in sound environments.

P1-5

Spatiotemporal occurrence patterns of northern bottlenose whales within the Nordic Seas based upon passive acoustic monitoring (PAM)

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Northern bottlenose whales (Hyperoodon ampullatus) are echolocating, deep-diving beaked whales found primarily in arctic and sub-arctic offshore waters. The eastern North Atlantic population has been suggested to undergo seasonal north-south migrations, however, previous whaling data and more recent sighting surveys insufficiently covered the winter months. We address this data gap via passive acoustic monitoring. Bottom-moored hydrophones were deployed at three locations in the Nordic Seas: off Jan Mayen (Norway; 2015-2017), north-east Iceland and east Iceland (2020-2023). Automated click detection was conducted based on species-specific click characteristics. Detector performance was manually evaluated using a ~10% subset from each deployment (n=8) and used to calculate the weekly proportion of time containing northern bottlenose whale clicks. Generalized additive mixed models were used to investigate environmental variables associated with prey availability which might drive distribution changes. Results revealed near year-round presence of bottlenose whales in the Nordic Seas with a gradual northward shift in spring between Iceland and Jan Mayen. The lowest numbers of detections occurred from July to September, contradicting the long-standing hypothesis of north-south migrations for overwintering at lower latitudes. The observed seasonal occurrence patterns were stable across years and linked to variations in sea surface height, indicative of eddy formation. We interpreted this as support of a prey-driven distribution, as eddies are characteristic of spawning grounds of bottlenose whale's main squid prey, Gonatus fabricii. The observed occurrence patterns should be considered in the planning of future anthropogenic activities in these waters to avoid habitat degradation and reduce stressors on this species.

P1-6

Analyzing bearded seal vocalizations: assessing automatic detection accuracy and distribution analysis

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Bearded seals in the Canadian Arctic face growing pressure from climate change, which is rapidly reducing the sea ice they depend on for habitat. It is important to study and build a baseline understanding of this species, but this is challenging due to the Arctic's extremely cold temperatures and low light levels throughout the year. Passive acoustic monitoring is a powerful tool for monitoring marine mammals in the remote Arctic environment, as underwater acoustic recorders can be deployed for long periods. However, once the recorders are recovered, there can be terabytes of data that must be searched to identify vocalizations at the species level. To aid in the efficient processing of large quantities of acoustic data, our team has been developing a deep-learning detector for bearded seal vocalizations. The goal of my Master's research is to test the efficiency of this detector and to apply it across 10 years of data from up to 11 different sites within the western Canadian Arctic to study the spatial and temporal distribution of bearded seals. In this presentation, I will share preliminary results on the detector's performance and initial findings of bearded seal presence across the study sites. This research will contribute to identifying important habitats for bearded seals within this region, helping to fill a critical knowledge gap for the species.

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P1-7

A biologist's guide to reproducibly integrate AI into passive acoustic population monitoring studies

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With accelerating global biodiversity loss, improving population and biodiversity monitoring is critical. Long-term data is much needed, but collection is resource-intensive, making rapid surveys and automated methods essential. The vast volume of data generated by passive recording units poses challenges in management and analysis. AI offers a powerful tool for processing large datasets efficiently, and the increasing availability of off-the-shelf AI software has made these methods more accessible to biologists, many of whom lack computational backgrounds. Despite these advancements in tools, ecological research also still faces a reproducibility crisis, with many studies failing to replicate results due to the lack of standardized computational workflows. While AI can accelerate data processing, improper implementation can lead to systematic errors. Passive acoustic monitoring often requires intermediate steps such as manually processing many hours of recordings. Observer bias, noisy biological data, and variability in recording systems challenge model accuracy, particularly under fluctuating environmental recording conditions. Here, we propose a framework for integrating AI into passive acoustic monitoring with an emphasis on reproducible workflows at the annotation stage. We outline best practices to improve model implementation and ensure robust, scalable monitoring. Using a marine soniferous fish species, the Atlantic cod (Gadus morhua), as a case study, we outline how advancing system-specific knowledge in evolutionary ecology and acoustic communication has the potential to refine AI tools beyond presence/absence detection. By standardizing AI integration, we aim to enhance the reliability and effectiveness of biodiversity monitoring, ensuring AI-driven tools contribute meaningfully to conservation and ecological research.

P1-8

How to count a click: Harbour porpoise behaviour and sensor deployment affect the probability of detecting echolocation clicks, click trains and detection positive time units

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The visually-elusive harbour porpoise (Phocoena phocoena) continuously produces stereotypical narrow-band high-frequency echolocation clicks for foraging, navigation, and communication. This makes passive acoustic monitoring (PAM) an attractive option for population assessment. To estimate the density of animals in a specific area it is essential to know how the probability of detecting animals changes with distance from the PAM sensor. This probability depends on many factors including sea bottom topography, PAM sensor deployment geometry and animal behaviour. In acoustic analysis, click detections can be expressed in various units, which can affect results, e.g., individual clicks, consecutive click sequences ('click trains') or time units that contain at least one click (e.g. 'detection positive minutes'). We explore these effects using data from tag deployments on 11 harbour porpoises in Danish waters, which yielded fine-scale acoustic and movement behaviour for up to 54 hours per tag. We simulated a grid of PAM sensors around each track, used a spherical spreading model for sound propagation and simulated the porpoise sound beam to estimate the effect of deployment geometry, environment and behaviour on detectability for different units of acoustic analysis. This allows us to assess which variables most affect the probability of acoustically detecting an animal, and to examine which units of analysis are most robust to changes in these variables. Ultimately, we aim to enable synthesis of porpoise PAM data collected across many studies in the North Sea to give an overall picture of porpoise trends over space and time.

P1-9

Passive acoustic monitoring of bird communities in urban public parks: A case study from the Dan metropolitan area, Israel

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Israel is a small, densely populated country with high avian diversity. Rapid urbanization processes lead to habitat degradation and fragmentation, making urban green spaces increasingly important for both birds and humans. We used passive acoustic monitoring to characterize avian communities in urban parks across the Dan Metropolitan Area, Israel's largest metropolitan region. We deployed AudioMoth recorders in 30 parks across 10 cities during March-April 2023. A custom BirdNET-GUI Analyzer model was trained to identify 35 bird species, including both residents and migrants. The custom model achieved a precision of 0.96, a recall of 0.47 and F1 score of 0.63. We examined acoustic monitoring results in relation to vegetation height distribution characteristics within parks using LiDAR technology. This integrated approach aimed to provide continuous and reliable monitoring of avian diversity in urban areas of the country. Understanding urban acoustic landscapes is essential for biodiversity conservation and human well-being. Preliminary results suggest that acoustic monitoring can effectively characterize bird community composition and that park vegetation structure affects avian species richness.

P1-10

From data collection to conservation: The strengths of large-scale collaborative passive acoustic monitoring

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Recent decades have seen a substantial increase in the collection of passive acoustic monitoring (PAM) data, driven by rapid technological advancements in recording systems, storage capabilities, and automatic detection and classification of signals. The expansion in PAM enables researchers to address biological questions on a larger scale than ever before, such as determining spatiotemporal patterns of abundance, movement patterns over entire habitats, and changes in acoustic behavior linked to environmental drivers. Large-scale PAM studies rely on the compilation of PAM data across research teams, thereby increasing effort, coverage, and statistical power, allowing for the quantifying of uncertainties and robust inference. For example, decades of collaborative PAM research have come together to allow near-real-time monitoring of North Atlantic right whales (Eubalaena glacialis) in the United States, providing a much-needed conservation tool for the critically endangered species. PAM can also be effectively used terrestrially, such as in the case of large-scale monitoring of bats across Finland which identified a migratory shift in an endangered species. Here, we advocate for more large-scale, collaborative PAM research to more effectively monitor species over biologically relevant scales. As part of the EU Horizon project, Digital Twin Ocean, we present a use case in which PAM data from diverse contributors allows us to model harbor porpoise movement, behavior, and potential disturbance from future offshore energy development across European waters.

P1-11

Monitoring realities: The application of imperfect data in conservation

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Using acoustic recorders to monitor wildlife is only valuable if we can assign sounds to species or species groups. Species groups often occupy different acoustic niches (e.g. large sperm whales produce lower frequency echolocation clicks than the small harbor porpoise); however, there can be overlap between species which both share an acoustic niche and produce similar sounds. Echolocation is typically a short pulse with a broad frequency range that is comparable among similar sized species. In most long-term passive acoustic monitoring (PAM) surveys, the animals recorded are free roaming, meaning that the angle from the acoustic midline of the animal to the recorder (aspect) is unknown. Due to the narrow beam, most clicks are recorded off-axis and hence with distorted waveform and frequency spectrum. The development of automated species detectors necessitates that clicks contain stable, distinct characteristics. For example, most beaked whales produce a species-specific, frequency-modulated click that are largely unaffected by the aspect angle. Moreover, the off-axis clicks of certain dolphin species contain spectral peaks/notches that are suitable for automated classification. We present how the relative density can be calculated from PAM data, as exemplified with Ziphius cavirostris, or how the distribution of a highly cryptic species can be better understood, as with Mesoplodon carlhubbsi, off the US west coast. Additionally, we present the results from the first full year of monitoring for Lagenorhynchus albirostris in Skagerrak waters. When a click type is confidently linked with a species or group, PAM data better informs mitigation and species conservation efforts.

P1-12

Conducting a streamside sonata: the impacts of restoration and invasion on urban stream soundscapes

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When implemented, post-restoration faunal monitoring efforts largely focus on indicator taxa. Similarly, studies evaluating invasive species' impacts on native wildlife often evaluate the effects of a singular invader on a singular native species, though most systems have multiple invaders. Therefore, we used passive acoustic monitoring and acoustic indices to monitor the soundscapes of urban streams impacted by restoration and invasive plants across 20 paired restored and unrestored stream reaches (n = 40) in Maryland's Chesapeake Bay Watershed. Acoustic recording units were placed near the midpoint of 100-m vegetation sampling reaches, within 5 m from the streambank, and set to record 15 minutes of every hour, every day for one year. We employed generalized additive mixed models to analyze five commonly used acoustic indices as response variables. Restoration was associated with more intense sound events and increased biotic noise relative to human noise, while invasion was associated with more uneven soundscapes and more human noise. Restored stream reaches may help to create mosaic habitats across the urban landscape that could lead to more intense sound events. Invasive species are often associated with urbanization and thus it was unsurprising that more invaded reaches were characterized by increased human noise and more uneven soundscapes, potentially due to traffic noise. Our research is one of the first studies to relate invasive plants to soundscapes and informs our understanding of how restoration and invasion impact audible species' responses to environmental conditions.

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P1-13

Few-shot transfer learning enables robust acoustic monitoring of wildlife communities at the landscape scale

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Pre-trained classifiers for bioacoustic species detection have emerged as accessible tools for conservation and automated biodiversity monitoring at scale, yet can fail to generalize in applied scenarios due to novel acoustic context (e.g., abiotic noise, concurrent vocalizations). This can ultimately preclude reliable ecological inference from passive acoustic surveys. Although relatively underexplored in acoustic monitoring, transfer learning may help overcome the limitations of existing pre-trained classifiers by enhancing predictive performance and detection capabilities while avoiding the time, resources, and expertise required to develop entirely new models. We demonstrate the use of few-shot transfer learning to adapt a pre-trained BirdNET source model to an applied monitoring domain with minimal local training data, enabling robust surveys of wildlife communities across the landscape. In a case study of avian biodiversity monitoring across a temperate rainforest landscape in Washington, United States, the model achieved a mean precision-recall AUC of 0.94 at the audio segment level and a mean accuracy of 92% at the site-use level across trained species classes, an increase of 0.16 and 13% over the pre-trained source model, respectively. These improvements increased the probability of individual species detection and produced more accurate estimates of local species richness. Superior performance was achieved with an average of only 8 local training examples per species. Training novel classes for other signals of interest (e.g., insects, aircraft, logging, rain) reduced confusion rates and mitigated the impact of the local environment on performance. We provide guidelines for practitioners to apply transfer learning to other environments and taxa.

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P1-14

Combining acoustic and visual surveys to estimate Indian Ocean humpback dolphin populations along the Sindhudurg Coast, India

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Conventional methods for estimating cetacean populations like visual distance sampling, assume perfect trackline detectability. However, these surveys face limitations when applied to coastal species inhabiting turbid waters. Indian Ocean humpback dolphins off the west coast of India are one such example. Together with environmental and weather conditions affecting detectability, visual surveys in India are severely constrained by lack of trained personnel. Our study integrates acoustic-visual detections in a mark-recapture distance sampling (MRDS) framework to calculate a correction factor for visual trackline detectability. These data are then used to estimate humpback dolphin densities along a 376 km^2 area off the Sindhudurg coast, India. Acoustic and visual line transect surveys were conducted between January 2020 and March 2023. Acoustic data were continuously collected using a four-channel towed array while three observers scanned the water surface for visual cues. Visual and acoustic detections were used to calculate perpendicular distances to animals to estimate trackline detectability of the visual platform using a MRDS framework. The independent observer configuration with a hazard-rate detection function estimated a visual trackline detection probability of 0.7 (CV = 9.8%). The model estimated an abundance to 191 individuals (CV = 28%). We further discuss these results in the context of improving visual distance sampling estimates using the calculated correction factor. Our study provides a framework for integrating acoustic and visual data to estimate population size for a data deficient species. These findings contribute towards developing efficient monitoring protocols especially for regions where data collection is hampered by limited resources.

P1-15

World Oceans Passive Acoustic Monitoring (WOPAM) Project: How can a day of global recording across the world's oceans support scientific collaboration, conservation partnerships and creative use of sound to engage action.

Isla Keesje Davidson, Aaron Rice, Miles Parsons, Steve Simpson (and 200 + WOPAM collaborators).

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Earth's aquatic environments include a diversity of soniferous species, many still to be identified, within soundscapes reflecting ecosystem health and condition. Most are increasingly dominated by anthropogenic noise. As passive acoustic monitoring (PAM) becomes increasingly accessible to scientists, conservationists and the general public, there is real opportunity to both explore and promote the importance of aquatic sound. Coinciding with the United Nations World Oceans Day on June 8, WOPAM (World Oceans Passive Acoustic Monitoring) project started in 2023, galvanizing efforts to collectively record the sounds of the ocean worldwide. As one of the first globally synchronized efforts, over 200 international collaborators have come together since then to record the soundscapes of over 400 different global locations every World Oceans Day for WOPAM. WOPAM is shaped by its collaborative global community and involves bioacousticians, artists and conservation practitioners. We will showcase how WOPAM aims to use its acoustic datasets and initial findings to support research interests identified within its community, as well as how it can facilitate knowledge exchange with local partners engaged in conservation action. WOPAM will contribute to catalogues of global acoustic diversity, help identify previously unknown sounds and collate acoustic trainings approaches. To encourage international multidisciplinary participation in aquatic bioacoustics on a global scale, it was involved in the UNs Oceans Conference and facilitated broader public engagement projects with aquariums. By sharing expertise, conservation partners are working to incorporate PAM into biodiversity monitoring efforts, establishing baselines in under-represented areas, and monitor anthropogenic impacts over time.

P1-16

The audio:tracker- realtime acoustic monitoring of bats and birds on autonomous sensors in the field

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Passive acoustic monitoring has become one of the most important biodiversity monitoring tools. However, the large amounts of recorded data that have to be analysed retrospectively prevent acoustic monitoring from being used on a large scale and over long periods of time. Permanent monitoring installations over several years to record population trends and phenology would generate amounts of data that are impossible to handle even if artificial intelligence such as BirNet is used for the subsequent processing of the collected data. To solve this problem, trackIT systems, in collaboration with the University of Marburg, has developed a system that uses small and powerful artificial intelligence to recognise the calls of bats and birds in real time on a raspberry pi-based autonomous sensor. The incoming audio stream is permanently analysed and species recognition takes place before an audio file is stored. All detections are transmitted to a remote server in almost real time and visualised there. This procedure enables a pre-configured decision about the species detections and the quantity of files to be stored for validation and thus a massive reduction in the amount of data generated. The system has been deployed in several locations throughout Europe including the Skagen Bird Observatory and the Zugspitze, Germany's highest mountain. To our knowledge, there is currently no system that enables the parallel detection of bats and birds in real time and allows decisions to be made about the amount of data to be stored based on the detections.

P1-17

SoundScapeExplorer: a free tool based on unsupervised learning for studying and monitoring natural soundscapes

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Through concrete examples, we present a free, open-source software package dedicated to the passive acoustic monitoring of natural soundscapes. Resulting from a collaboration between acousticians, biologists, machine learning specialists, and software developers, this tool has been designed with both performance and ease of use in mind, while maintaining the objective of extracting significant biological insights. It is based on unsupervised learning approaches, allowing the processing of large datasets without extensive manual labeling. It supports multiple neural networks and offers various types of result analysis (e.g., temporal analysis, cluster similarity). Its graphical interface facilitates the configuration of calculations, as well as the visualization and export of results.

P1-18

Structure and sound pressure level of calls of rails (*Rallidae*) - consequences for the propagation of signals

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The increasing use of acoustic recording methods for ecological research and environmental monitoring requires an objective assessment of the recording system's detection radius. Detection radius can be determined by acquiring reproducible data on the loudness of vocalisations. Calibrated audio recordings of 4 species of rails (Rallus aquaticus, Porzana porzana, Zapornia parva, Crex crex) were made in the field. The distance between the recording system and calling individuals were determined by acoustic localisation with a linear microphone array. The sound pressure level calculations took frequency-dependent atmospheric absorption, spherical spreading and background noise into account. Atmospheric sound absorption was calculated according to ISO 1996-1. In order to account for the differing contributions of the individual frequency components of the sound signals, the absorption coefficient for the centre frequencies of several octave bands was calculated. For each combination of distance, temperature and humidity, a set of FIR filters was created and applied before calculating sound pressure level (SPL). We calculated the sound pressure level as peak equivalent SPL, peak level and as rms. The acoustic structure of a signal, especially the distribution of energy within frequency bands, has a significant influence on its range. To increase SPL measurement reproducibility, we recommend either storing measured signals together with a calibration tone or using recording devices with a fixed dynamic range. We recommend building up a reference library of calibrated sound recordings, including information on meteorological conditions and distance estimation methodologies.

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P1-19

Beyond the soundscape: Challenges in using acoustic indices in Indian evergreen forests

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Rapid assessment programs using soundscape are widely used for biodiversity assessment and monitoring. Analyzing soundscapes using acoustic indices can also indicate changes in presence and abundance of species and overall ecosystem health. We aimed to assess the acoustic diversity in the tropical evergreen forests of the biodiversity hotspots regions of India using acoustic indices. Ambient acoustic data was collected from field surveys in the Western Ghats, Northeast India (2015 - 2020), Western Himalaya, and Central India (2023 - 24) using hand-held ultrasound recorders with a sampling rate of 500 kHz. The sampling area was classified into three ecosystems based on the land cover type- primary, secondary, and disturbed forests. Alpha indices- Acoustic Complexity Index, Bioacoustics Index, and Acoustic Diversity Index were calculated using R packages 'soundecology' and 'seewave'. Alpha indices were calculated separately for 1-10kHz, and 10-100kHz ranges with different frequency bins, 1kHz for the former and 10kHz for the latter, as the 1-10 kHz range was inhabited by narrowband callers and 10-100 kHz by broadband. Yet, acoustic indices were neither predictive of the species richness nor change in the ecosystem. Our findings indicate that acoustic indices should be used cautiously when the acoustic community is dominated by broadband and ultrasonic callers. Biodiversity assessment using automatic identification in conjunction with in-person field surveys in the future would provide reliable information on acoustic community richness and assemblages.

P1-20

Sound source localization of Japanese tree frog calls using time difference of arrival

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The anthropogenic introduction of translocated or non-native species poses concerns regarding the ecological disturbance of native species. For the conservation of native species and management of alien species, it is essential to monitor spatial distributions and population densities for understanding the pathways of invasion and its spread. Acoustic detection systems capture critical spatial biological parameters during field studies of vocally communicative amphibian population. However, localizing an individual frog is difficult because frog calls consist of multiple individuals. In this study, we conducted a sound source localization model using Time Difference of Arrival (TODA) of frog calls based on acoustic recording from a controlled experimental field where frogs were restricted to a limited area than in natural settings. Two recorders with stereo microphones were used to record at five-minute intervals. In the experimental field, Japanese tree frog (Dryophytes japonicus) inhabit individual 40 pods. A virtual experimental field, 6 m × 30 m, was constructed for TDOA-based machine learning. One-second pseudo-frog calls were placed at 1 m intervals to obtain a theoretical TDOA, and the TDOA of all possible pairs of four microphones was simulated. Through cross-correlation of each microphones pairs every second, the simulated TDOA was used to develop a data-driven model for the sound source localization. Our findings confirmed pressure differentials between recording devices occasionally displaced the apparent sound origin beyond the water bottle's confines, despite its actual inside. Future work will construct a precise data-driven pressure compensation model, which will lead to practical application in actual paddy field environments.

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P1-21

Variability in inshore soundscapes on the southeast coast of South Africa

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In South Africa, underwater noise pollution has been acknowledged as an emerging threat to marine environment and is expected to increase with the development of the country's ocean economy. The impacts associated with ocean economy activities have not yet been quantified for South Africa's marine environment. Ensuring appropriate governance and application of effective mitigation measures to protect marine biodiversity will be contingent on research to characterise marine soundscapes and investigate the impacts of anthropogenic noise on marine biodiversity. This study examines the soundscapes of two inshore Marine Protected Areas and two unprotected locations on the southeast coast of South Africa, using PAM. Soundscape data were collected during February–May 2024 using AAR's. To visualize the acoustic recordings in frequency and time domain, median-based long-term spectrograms were used, with spectral variation analysis to assess the spectral–temporal variations in soundscapes of each site. To separate sound sources and investigate the diurnal patterns of each sound source in the soundscape of each site, non-negative matrix factorisation modelling was performed, as well as soundscape divergence analysis based on Uniform Manifold Approximation and Projection. Preliminary results revealed significant divergence among sites, with all sites exhibiting high biophony, characterized by fish choruses and snapping shrimps that showed a clear diurnal pattern. Vessel noise was the main source of anthropogenic noise contributing to the soundscape of each location. Other sound sources contributing to the soundscape include dolphin whistles, waves and rainfall. The findings of this study shows that PAM is an effective tool for long-term soundscape monitoring.

P1-22

Enhancing baleen whale conservation through advancements in distributed acoustic sensing

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Conservation bioacoustics transforms sound into ecological insights, empowering conservation efforts at diverse scales. Yet, traditional acoustic monitoring methods provide limited spatial resolution on whale vocalizations, constraining research on habitat use, movement patterns, and population estimates. Distributed Acoustic Sensing (DAS) is emerging as a revolutionary solution, repurposing submarine fiber optic cables into vast, high-density listening arrays. Instrumented from land, DAS can detect and track low-frequency baleen whales at regional scales with nearreal-time data delivery—a breakthrough for dynamic, responsive marine ecosystem management. To fully realize its potential, DAS requires rigorous calibration and performance validation. This talk explores how DAS converts strain measurements into Pascals while addressing higher frequency capabilities across diverse ocean environments—including the North Atlantic, Northeast Pacific, Mediterranean, and Chukchi Sea. At low frequencies, a universal strain-to-Pascal conversion factor was identified, but detection ranges were limited by the self-noise of commercially available interrogators. These findings highlight the need for reinforced interdisciplinary collaboration between ecologists, engineers, and industry innovators. By uniting advanced signal processing, engineering, and ecological monitoring, DAS offers an unprecedented opportunity to reshape baleen whale conservation. With scalable, near-real-time data delivery at regional scales, DAS bridges cutting-edge technology and actionable insights, paving the way for dynamic, data-driven marine mammal management. This transformative tool holds the potential to redefine how we monitor and protect whales in a rapidly changing ocean.

P1-23

Call for collaboration: Sharing European bat sound data for open bioacoustics AI

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Bats are key players in ecosystem health, yet their nocturnal, aerial, and often cryptic behaviour makes them particularly difficult to monitor. This monitoring gap limits our ability to detect population changes, including responses to threats such as habitat loss and insect decline. Improving how we track bat populations is therefore essential for effective conservation planning. Advances in technology are revolutionizing how we monitor biodiversity, making it easier, and more cost-effective to track species remotely. As part of the MAMBO project (Modern Approaches to the Monitoring of Biodiversity), we are developing Artificial Intelligence (AI) tools that use sound recordings to support large-scale, remote monitoring of bats, especially those native to Europe. By combining sound recognition with deep learning, we aim to enable faster, more accurate species detection, contributing to the EU Biodiversity Strategy for 2030. A key requirement for training robust, generalizable AI models is access to high-quality, labelled bat audio recordings from a range of species and habitats. We are therefore inviting researchers and institutions to contribute annotated bat audio data to Xeno-canto (www.xeno-canto.org), a trusted and widely used open-access sound repository. By pooling data in a public, community-driven repository, we can support the development of transparent, reusable tools for biodiversity monitoring. This collaborative effort will strengthen research networks, facilitate knowledge exchange, and contribute to long-term conservation goals through shared infrastructure.

P1-24

ESO: An evolutionary algorithm for efficient model reduction via automated frequency band selection in passive acoustic monitoring

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Passive Acoustic Monitoring (PAM) is a vital tool for wildlife monitoring, and deep learning, particularly convolutional neural networks (CNNs), has become the standard approach for developing bioacoustics classifiers. However, real-time classification remains challenging due to the high computational complexity of these models and the need for resource-intensive preprocessing steps, such as downsampling. One approach to reducing model size and computational complexity is decreasing input dimensions, specifically by reducing spectrogram size—the visual representation of sound. We introduce Evolutionary Spectrogram Optimization (ESO), an evolutionary algorithm that automatically identifies and extracts narrow frequency bands from spectrograms that are informative enough to allow CNN-based species detection. ESO optimises the selection and size of these bands, while maximising classification accuracy and minimising the number of CNN parameters. It eliminates preprocessing steps such low-pass filtering and downsampling, optimising real-time processing. We evaluated ESO on three species: the Pin-tailed whydah, Thyolo alethe, and Hainan gibbon. Compared to a baseline CNN trained on full spectrograms with preprocessing, ESO reduces spectrogram size by up to 70%, decreases CNN parameters by 78%, and improves F1-score by up to 7%. As a key example, ESO increases inference speed by 45%, reduces peak memory usage by 47%, and lowers energy consumption by 56% for the Pin-tailed whydah dataset. By reducing model complexity while maintaining high classification accuracy, ESO is a promising solution for real-time PAM applications. It enhances computational efficiency by reducing inference time, memory usage, and power consumption, making the method well-suited for onboard deployment.

P1-25

Sound of the effect: using a passive acoustic monitoring approach to detect sub-lethal behavioural responses to a Neonicotinoid

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Chemically induced behaviour modifications due to exposure to pesticides are believed to be a main driver behind declining insect populations. Soil-dwelling insect populations might be at the most risk, not only as soils serve as strong recipients of pesticides but also as we currently lack non-invasive methods to detect in situ behavioural responses from neonicotinoids on soil macrofauna. In this study, we hypothesise that a small shift in behaviour for the common redwood ant (Formica rufa) due to sub-lethal exposure to a common neonicotinoid (imidacloprid) can be registered by a passive acoustics monitoring system. In a series of independent field experiments, we show that both simple acoustic proxies, as well as more advanced AI analyses of acoustic recordings, can indeed be used to detect sub-lethal behaviour shifts (i.e. increased activity) induced by the pesticide. The acoustic proxy for activity (peak power) revealed a short-term (<6 hours) response to body burdens of 10.8 ng imidacloprid g-1 ant after 24 hours. This short-term response was supported by similar trends for cholinesterase activity, suggesting that ants were able to recover after 6 hours. We conclude that acoustic information from forested seengs can be used for early detection effects of neonicotinoid exposure on a main group of soil fauna (ants).

P1-26

Are fear and aggressive nonverbal vocalizations similar across diverse cultures?

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Human non-verbal vocalizations such as fear screams and aggressive roars have stereotyped acoustic forms that can be predicted based on their communicative and adaptive functions. Following this form-function theoretical framework, fear vocalisations are characterised as high-pitched and tonal to communicate alarm, whereas aggressive vocalisations are expected to be low-pitched and harsh to communicate threat. However, little is known about how and why such vocalisations vary across human societies. Most research on human non-verbal vocalisations has been limited to WEIRD (Western, Educated, Industrialized, Rich, and Democratic) populations, leading to scientific ethnocentrism. Here, to examine the generalisability of these form-function relationships, we recorded individuals producing fear and aggressive non-verbal vocalisations in three highly distinct human populations: the Darkhad and Hadza groups (two non-WEIRD populations in Mongolia and Tanzania), and French urban dwellers. We then measured several acoustic parameters known to be biologically and perceptually relevant in human non-verbal vocal communication to assess the ostensible universality of the acoustic structure of these vocalisations. Using multilevel Bayesian mixed models, we found that previously established form-function predictions were consistent only for the French group, suggesting that the Darkhad and Hadza groups do not rely on exactly the same acoustic parameters to communicate these emotional states. Furthermore, a random forest classification successfully discriminated between fear and aggressive vocalisations from the three different groups, suggesting the existence of a group signature. These results contribute to a deeper understanding of the acoustic space and evolved functions of the human vocal repertoire.

P1-27

Strong sexual dimorphism in the songs of Galapagos yellow warblers

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Early research on bird song suffered from a male bias. In the last two decades, however, an increased effort to address these biases in song research revealed the abundance of singing females even in well-studied species. Female song in the Yellow Warbler (Setophaga petechia, one such well-studied species) has previously been described to be a rare occurrence as solo songs, but in the Galapagos subspecies (Setophaga petechia aureola), we recently discovered that female song is common and sung as duets with the paired male as well as solos. Here, we investigated acoustic differences between male and female songs from targeted recordings of more than 40 pairs. We found that songs of females had a narrower fundamental frequency range and more energy in overtones than those of males. Unlike the song of male yellow warblers, the female song often contained biphonation, subharmonics, and deterministic chaos. We discuss these differences from the perspective of source-filter theory and the natural history of yellow warblers.

P1-28

Bats approaching prey adjust the SPL ratio of pulse and echo for optimal ranging

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Bats lower their emission SPL when approaching a target, possibly to match the operational range of delay-tuned neurons for optimal range processing. This implies that the reduction rate of emission SPL should differ between bats with low and high amplitude search signals. In this study, we recorded echolocation signals from four low-amplitude Myotis myotis and two high-amplitude Eptesicus fuscus using onboard microphones during approaches of mealworms and reflectors of varying target strength. We analyzed the emission and echo rms SPLs at the bats' ears and determined the reduction rate of the emission SPL and increase rate of the echo SPL. M. myotis reduced their emission SPL from approximately 90 dB to 75–70 dB, resulting in a reduction rate of about 3.2 dB/hd. Echo SPLs increased from around 20 dB to 40 dB during buzz 1. Conversely, E. fuscus decreased their emission SPL from 106 dB to 95–88 dB, corresponding to reduction rates of 11.3, 10.5, and 6.7 dB/hd for different targets, maintaining echo levels near 40 dB throughout the approach to all targets. These findings suggest that bats with low emission SPL face the challenge of increasing echo SPLs for optimal processing at the end of the approach, resulting in only slight SPL reductions. In contrast, high-amplitude bats receive near-optimal echo SPLs early and must strongly reduce their emission SPL to maintain the optimal pulse-echo ratio. Overall, our results support the hypothesis that bats adjust their SPLs to keep the pulse-echo ratio within bounds for optimal range processing during approach.

P1-29

Vocal scaling in wild bats: do larger species call lower, louder, and less often in the same auditory environment?

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Echolocating bats use sound to navigate and forage in highly dynamic environments and display a high degree of auditory and vocal diversity, presumably to tailor sensory inputs to informational needs. Anatomy and biomechanics are hypothesized to structure this call diversity, where body size scaling principles posit that bigger bats should call lower, louder, and less often. However, due to the unique way that bats use ultrasound, these patterns are likely also shaped by individual foraging specialisations and the physics of sound propagation. Here, we test the hypotheses that 1) vocal diversity follows common body size scaling in bats, but that 2) scaling coefficients are affected by specific foraging specialisations of individual species and the physical circumstances of their sound production. We did so by recording calls from seven bat species presenting four guilds during their nightly emergence through a cave tunnel with a microphone array, therefore ensuring that all species faced the same sensory scene, physical conditions and behavioural task. Emitted frequency scaled negatively with body size as predicted from allometric scaling. However, call intensity unexpectedly also scaled negatively with body size (i.e. smaller bats called louder), while call rate did not scale with body size as would be predicted by the coupling between wingbeats and calling. We conclude that the diversity of vocal parameters is only in part predicted by body size, but that diverse foraging strategies and the high ultrasonic absorption in air play important roles in shaping the vocal diversity throughout echolocating bats.

P1-30

Between a whine and a growl: Acoustic and perceptual characterisation of dog moans

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Dog vocal behaviour is versatile, especially due to intermediate calls. While barks, growls, and whines have been studied, more elusive calls like moans are largely unexplored. In wolves, moans are distinct calls that acoustically fall between the whine and growl, but remain unstudied in dogs. Thus, in an online survey, participants (N=231) heard a random but balanced sample of ten dog call types, including whines, growls, and moans. In a forced single-choice task, they labelled calls with literature-based terms, then positioned growls, moans, and whines on a visual analogue scale of similarity.

In the subset of growls, moans and whines, fuzzy clustering identified three clusters based on call length, pitch, and tonality. Whines and growls mostly fell into two clusters, while moans were spread across all three, their majority fell into the third. PDFA and GLMM confirmed significant differences among these calls: whines were shorter and higher-pitched than growls and moans, while growls were noisier than whines and moans. Naïve listeners identified call types above chance, but often misclassified moans, which were also rated as intermediate between whines and growls. Both across all sounds and within moans, shorter, higher-pitched and tonal calls were rated closer to whines, while longer, lower and noisier calls were closer to growls. Overall, findings suggest that dogs' moans, growls, and whines are distinguishable acoustically and perceptually, but the boundaries between categories are not strict, forming a graded vocal continuum within a possible combined vocalisation type.

P1-31

Mouth movements during high call rates in vesper bats

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Bats dynamically adapt echolocation calls context dependent through modification of intensity, frequency, duration, pulse interval and directionality. Their vocal system is highly optimized for rapid sound production utilizing superfast muscles generating call rates >160Hz. Contrary, much slower facial muscles dictate the vocal tract filtering properties and as such influence emission intensity, frequency and directionality. Previous studies suggest fine, call-by-call, control of the emitter size and thus a change in call directionality to reduce clutter echoes and increase forward sensitivity. Given the slower time scale of vocal tract modifications, our hypothesis predicted that such fine control is limited by the maximal contraction speed of jaw muscles and thus echolocating bats would stop mouth movements at high call rates. To test this, we trained individuals of two vesper bat species Myotis Nattereri and M. daubentonii to hunt prey while recording their behaviour with stereo high-speed videography and mapped their emission using a multi-microphone array. We used an artificial neural net to markerless track mouth movements and correlated those to measured acoustic parameters. Preliminary data shows that prey approaching bats increase the mouth gape with increasing call rate. However, as call rate increase the bats do not close their mouth fully between emissions, suggesting a muscle-speed limitation. We believe bats tailor mouth movements and call emission for a dynamically optimized sonar scene and that the movements are constrained by muscle speed and fatigue limits.

P1-32

Syrinx muscle territories: Spatial organisation of the syringeal motor pool for the neural control of birdsong

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Birdsong is an excellent model system for understanding acquisition and maintenance of complex fine motor skills, but we know surprisingly little about how the motor pool controlling the syrinx is organized. Syringeal muscles of the zebra finch (Taeniopygia guttata) possess extremely small motor units (MUs) with 13-17% of motorneurons innervating a single muscle fibre. However, the remaining majority of MUs are larger and their associated muscle fibres must be distributed over the muscle in a so-called territory. The spatial organisation of these territories determines how muscle force is transmitted but is unknown for most muscles, including syrinx muscles, due to methodological constraints. Here we measured MU distribution in living intact syrinx muscle using spatiotemporal imaging of intracellular calcium concentration. We quantified the spatial distribution of 9 MUs, containing 3-9 muscle fibres across 4 specimens using nearest neighbour and quadrant analysis. MU territories spanned ~10±8% of the muscle area, overlapped spatially and contained many muscle fibres from other unidentified MUs. Although there may be a tendency to form subclusters, the spatial distribution of muscle fibres within the muscle territory was random, which supports earlier findings based on glycogen depletion. The random distribution of muscle fibres allows for a smooth force transmission to muscle attachment sites. The small size of syrinx muscles makes them an ideal model for general motor pool organisation.

P1-33

Bottlenose dolphin biosonar adaptations during task solving in shallow water

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Echolocation is a crucial sensory mechanism for odontocetes to navigate, forage and communicate in their marine environment. However, echolocation signals can be influenced by various environmental factors, including water depth. In this study, we investigate individual biosonar adaptations of bottlenose dolphins (Tursiops truncatus) to shallow water. With a two-alternative forced choice task, dolphins were asked to discriminate cylinders of distinct wall thicknesses in 4m and 0.5m water depth. We investigated how dolphins changed their source level and click rate in shallow compared to deep waters, by means of on-target acoustic recording synchronized with overview video recording. Source levels in deep conditions followed the expected trends of a 20log(R) decrease with range, but this tendency was not consistent in the shallow water trials where a more diverse trend was observed. Click rates followed our expectations, as dolphins used significantly more clicks to solve difficult tasks and more clicks in shallow compared to deep water. Our findings raise questions about the influence of water depth on dolphin biosonar performance and highlight individual flexibility in performing echolocation tasks.

P1-34

Stridulation-like behaviour in red wood ant (*Formica rufa*)

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Stridulation – production of sound by rubbing body parts together – is widespread in insects. While some ant species are known to stridulate, this is not the case for the Red Wood ant (Formica rufa group). In this study, we report that Red Wood ants generate rattling sounds resembling stridulation, being about 0.7 ± 0.2 second-long (mean ± standard deviation). The sounds were generated in our laboratory setting at a rate of about 0.6 productions per hour per individual. To investigate the use of these stridulation-like sounds we performed a playback experiment. Upon hearing the rattling sounds, ants displayed a reduced locomotory speed response, while ants exposed to silence did not. However, this reaction to stridulation-like sounds is ambiguous and its interpretation complex. Our current hypothesis is that stridulation-like sounds are used as a short-ranged means of keeping in touch with nestmates. Using X-ray micro-tomography, we describe a ridge structure located on the anterior margin of the pronotum that could produce stridulation-like sounds when paired with the leg as a scraper. Our exploratory study shows that the stridulation-like sound is faint and easily missed in behaviour assays despite playing a potential role in Red Wood ant communication.

P1-35

Doppler shifted echoes from the moving wings reveal tight coupling between sound emission and wing beat in Greater Horseshoe bats foraging in the wild

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Using sound and movement recording tags we studied the foraging behaviour of Greater Horseshoe bats in the wild. We describe the echolocation behaviour of bats searching for fluttering insects from perches and pursuing detected prey in short sally flights, of bats flying in permanent acoustic contact with the vegetatuon and capturing insects on the wing, and of bats flying over water. The recordings show clearly the emitted echolocation calls, the echoes returning form the environment, and at captures the echoes of fluttering prey. Notably, we also found negatively Doppler shifted echoes at the onset of single sounds and of the first sound in sound groups suggesting that the reflecting surface is moved backward in relation to the bat' head. We assume that the Doppler shifts are caused by the bats' wings. From former studies and from our synchronised accelerometer data we know that single sounds and the first sounds of groups start in the middle of an upstroke. Studies describing the kinematics of the flight of the Greater Horseshoe bat (Aldrige 1986) show that the wings are moved backwards in the upper part of the upstroke. Thus, the observed negative Doppler shifts are caused by the backward movement of the wings. They indicate a tight temporal coupling between wingbeat and sound emission. Only in the long terminal groups or buzzes emitted in landing bats and in bats catching an insect further negatively Doppler shifted echoes indicate further wingbeats. A few single sounds without wing echoes may indicate short gliding phases.

P1-36

Seeing sound: A new vision for acoustic perception

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Sound is a cornerstone of human experience, encoding essential information for communication, emotional expression, and environmental awareness. For deaf individuals, the inability to access this sonic world results in the loss of critical auditory cues, including non-verbal vocalizations, environmental and cultural sounds, which are vital for social bonding and emotional understanding. In the 1940s, spectrograms were tested to help deaf individuals decode speech visually but were abandoned due to technological constraints and speech complexity. Here, we revisit this concept, exploring whether modern spectrograms can visually convey non-speech acoustic signals. Our research investigates whether individuals can intuitively interpret spectrograms of salient acoustic signals, exploiting cross-modal correspondences where auditory features (e.g., roughness, rhythm) map to visual patterns. Across nine psychometric experiments, 840 naïve, typically-hearing participants evaluated acoustic features of non-verbal vocalizations (e.g., baby cries), environmental sounds (e.g., bird songs, animals' vocalisations), and anthropogenic sounds (e.g., alarms, music) under three conditions: auditory-only, spectrogram-only, or combined auditory-spectrogram presentation, with varying training levels—none, implicit (audio-spectrogram exposure), or explicit (tutorial with feedback). Participants rated low-level (rhythm, roughness, complexity) and high-level (distress, beauty, melody, pleasantness) dimensions on a continuous scale. Distress ratings for baby cries showed a 30% audio-visual correlation without training, rising to 70% with explicit training. Low-level dimensions exhibited stronger correlations (e.g., rhythm: r=0.65; roughness: r=0.60) than high-level ones (e.g., melody: r=0.40; beauty: r=0.35), with biologically or culturally salient sounds (e.g., music, bird songs) showing enhanced correspondence. These findings suggest a neural basis for cross-modal perception, highlighting spectrograms' potential as sensory substitution tools to bridge auditory gaps for deaf individuals.

P1-37

Sound localization behavior in the barking gecko (*Ptenopus garrulus*)

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The burrowing South African barking geckos, Ptenopus garrulus, have a robust calling behavior. Calls of male barking geckos serve to attract females, and in male-male competition by enabling males to maintain exclusive space surrounding male burrows. We investigated barking geckos in the field in Kuruman River Reserve, South Africa using auditory brainstem responses (ABR) to sound, laser vibrometry, anatomy and behavioral observations. Sound localization behavior was recorded on a video camera (under infrared illumination) directed at the burrow mouths of calling geckos to record behavioral responses to gecko calls played sequentially from eight Bluetooth speakers at 30˚ intervals. Orientation responses were quantified from the video recordings. Barking geckos are very sensitive to sound with peak sensitivity at 3 kHz. The ear is strongly directional from 2 to 4 kHz (the peak frequencies in their calls), where interaural transmission gain is close to 0 dB, showing strong interaural coupling. The geckos oriented to call playbacks with an accuracy of about 8°. The accuracy was highest from sounds from frontal angles and closer to 20° for the most lateral sound sources. The strong directionality of the barking gecko ear is similar to that of other lizards, while their uniquely robust sound communication behavior allows us to relate this directionality directly to behavior and processing of the communication calls. This first demonstration of sound localization in lizards should inform our understanding and modelling of directional sensitivity and neural processing of directional sound.

P1-38

Echolocating harbour porpoises (*Phocoena phocoena*) are capable of segregating close auditory streams during active target interception

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Echolocating animals are faced with the challenge of separating echoes of interest from extraneous background echoes, so-called clutter. Clutter from distant objects arrives later than echoes from nearby targets, and could elicit range ambiguity if the echolocator retransmits before echoes from its preceding click have returned. Additionally, if arriving within the same auditory integration window, clutter echoes may mask prey echoes. Reduced click rate or level may alleviate range ambiguities, as could auditory stream segregation; the latter is observed in bats. Toothed whales occupy inherently complex and cluttered auditory scenes, owing to long detection ranges and fast sound speeds in water, but how they accommodate auditory clutter remains unknown. We therefore asked how clutter influences target tracking strategy and interception in echolocating toothed whales and whether their auditory integration time imposes limitations. We presented two trained harbour porpoises (Phocoena phocoena) with artificial echoes generated using playbacks of their clicks. Artificial echoes were at fixed delays of 3 and 0.1ms during active target approaches, to evaluate the effects of clutter and the temporal limitations of auditory integration, respectively. The porpoises successfully intercepted targets, despite the presence of several echo streams. Moreover, porpoises did not adjust interclick-intervals or source levels to accommodate clutter, but buzzed longer when echoes occurred within their auditory integration window. This demonstrates that toothed whales can handle complex and tight auditory scenes, and suggests that they segregate auditory streams to maintain focus on targets of interest.

P1-39

Do birds have rhythm? Singing cadences of birds in a north-temperate forest

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Bird vocalizations are used for sexual courtship and territorial defense. But what aspects of a vocalization convey information? Numerous researchers have investigated the nuances of note structure, frequency modulation, and repertoire size, while comparatively few studies have examined the broader sequencing and timing of songs: rhythm. We define rhythm as the cadence of a bird vocalizations, or the regular onset and offset of melodic phrases. The pauses between these phrases create a cadence we call intervocalization intervals (IVI). To analyze IVIs, we used passive acoustic monitoring to record birds at 24 locations in Marsh-Billings-Rockefeller National Historical Park (Woodstock, Vermont, United States). For each location, we selected 20 postdawn recordings that were 10 minutes long, and from within those audio files, analyzed over 87,700 discrete vocalizations. All but two of 29 abundant species displayed nonrandom rhythmicity in more than 25% of the recordings. The median IVIs were strikingly similar within the Parulidae and Cardinalidae at about 15 to 20 seconds. Other clades were more variable. Rhythmicity of vocalizations could contribute to both intrasexual and intersexual communication. Regular cadence may make it easier for females to assess and track their mate, and for neighboring males to recognize territory boundaries. Additionally, regular cadence may distinguish auditory signals of a bird from those of other species and from other environment sounds. North-temperate forests are famously rich with bird songs in the early summer. It will be interesting to compare the vocalization rhythms of bird communities across other latitudes, biome types, seasons, and clades.

P1-41

Bats, tracks and tags: Lombard response and foraging success of the gleaning bat Trachops cirrhosus in noise.

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Many bats use echolocation, an active sensing system, to orient and forage in the dark. So-called 'gleaning' bats can also passively eavesdrop on prey-generated sounds to detect their prey, such as courtship songs (e.g. by frogs or by insects) or rustling sounds (e.g. by walking beetles). However, since echolocating bats constantly echolocate during flight, this raises the question of the relative importance of passive and active listening for hunting. To disentangle these respective roles in prey capture, we tested the gleaning bat Trachops cirrhosus in a prey-capture lab experiment during which we manipulated the bats' access to echolocation and to prey sounds with a robotic frog model and masking noise, recording the bats' echolocation behavior with bat- borne microphones. We found that noise which masks echolocation calls was a strong deterrent to bats which decreased the probability that they initiate prey attacks. The presence of passive cues increased motivation but not catching success. Further, our recordings suggest that fringed- lipped bats showed a strong Lombard response (increased call amplitude in noise) when noise masked echolocation call frequencies. In contrast, when noise masked the prey-generated cues, bats were less successful at detecting but more successful at catching prey. We did not detect any change in echolocation parameters. Our results are in accordance with previous studies of the Lombard effects in bats and other mammals; and suggest that Trachops cirrhosus rely on passive listening to localize the prey and initiate the hunting sequence, while they use echolocation for navigation and fine-tuning the attack.

P1-42

A silent strategy: quantifying acoustic camouflage at the community level with the mothbat arms race

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Acoustic camouflage in moths, i.e. the process of reducing the echo returned to a hunting bat, can play a crucial role in the biotic interaction between predator and prey. Despite its broad importance, this phenomenon has received relatively limited attention and only at the level of a few individual species. Here we explore the role of acoustic camouflage across entire communities of moths and bats, with the specific aim to test for local adaptation. By focusing on the temporal and spatial patterns of both bat and moth communities, we examine how acoustic camouflage relates to varying bat community dynamics, such as species composition, activity levels, and foraging strategies. Using a combination of field recordings and ensonification (quantifying acoustic camouflage), we assess the effectiveness of moths' acoustic defenses in different habitats (e.g., clutered versus open areas) throughout the season. We hypothesise that moths in areas with higher bat activity rely on higher levels of and more sophisticated ways of acoustic camouflage, potentially linked to shifts in bat echolocation behaviour. Insights from our large-scale data will enhance our understanding of predator-prey dynamics and the role of evolutionary pressures in shaping the acoustic make-up within ecological communities.

P1-43

Using Vaquita´s acoustic foraging behavior to establish priority conservation areas for its conservation

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The vaquita (Phocoena sinus) is a naturally rare endemic porpoise species restricted to the upper Gulf of California, Mexico. Its main survival threat is the incidental capture in poacher´s gillnets, with a rapid decrease in population size estimated in 2022 at 18 mature individuals. A population trend analysis was carried out using a grid of 46 automatic autonomous acoustic detectors (C-PODs) deployed within the vaquita refuge. The detectors registered the acoustic activity of the vaquitas during 2011 and 2018. Vaquitas emit Narrow-band high-frequency (NBHF) clicks with dominant frequencies between 128 and 139 kHz, with a width of 11 to 28 kHz, and a duration of 79-193 µs. Foraging trains were extracted and identified based on inter-click intervals, using criteria for harbor porpoises (Phocoena phocoena) foraging trains. The 132,951 vaquita´s click trains extracted were divided into 48,277 (or 34.7%) train buzzes with inter-click intervals between 10 and 2 ms; the other trains are yet to be classified. Our results indicate differential habitat use and pinpoint the most important vaquita foraging sites inside the vaquitas refuge. This finding may help us establish critical conservation areas for this highly endangered species.

P1-44

For fair (spectrogram) comparison you need to know your limits

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Spectrogram analysis is the most common approach to study animal vocalisations. However, under the same FFT (fast Fourier transform) parameters and sampling frequency, the same signal at different signalto- noise ratio (SNR) might look completely different. Thus, standard measurements of time and frequency on the signals without considering the SNR might lead to unreliable values. Comparison of spectrograms under these circumstances becomes unfair. Here, a methodology for a more reliable characterisations and ulterior comparison of animal vocalisation spectrograms is presented. The proposed technique is based on the 3D nature of the spectrogram, and the acknowledge that the amplitude and background of two given signals could hardly be the same.

P1-45

Fast & furious: Bottlenose dolphin biosonar during shallow water hunting

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Dolphins and other toothed whales use echolocation to detect and discriminate prey, but little is known about how free-ranging individuals use their biosonar in acoustically complex environments. In Shark Bay, Western Australia, Indo-Pacific bottlenose dolphins (Tursiops aduncus) engage in a variety of innovative foraging tactics to exploit different ecological niches. In our study, we investigated an individual foraging specialization performed by two adult females (mother and daughter), characterized by high-speed fish chases along the shoreline. We used drone-mounted videos with synchronized acoustic recordings to investigate how these dolphins adapt their swimming behavior and biosonar to facilitate prey capture in extremely shallow tidal waters (<1 m depth). Dolphins targeted mullet (Mugilidae) or golden trevally (Carangidae), using the shoreline as a barrier to herd prey, and chasing them with short sprints at up to 10 m.s-1. Source levels of dolphin echolocation clicks were lower than previously recorded from individuals in deeper waters, a potential strategy to avoid excess clutter levels from the substrate, sea surface and shoreline. We hypothesize that changes in body orientation, such as turning and swimming-on-side, improve maneuverability and acoustic detection of prey in shallow waters. Observations of largely overlapping foraging behaviors between the individuals studied hints at likely vertical mother-daughter transmission of the hunting technique, as has been shown in other individual foraging specializations in the Shark Bay dolphin population.

P1-46

Resonating priorities: Charting the future of bioacoustics

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Bioacoustic technologies are revolutionizing biodiversity monitoring by offering efficient, scalable, and non-invasive tools to identify species, understand behavior, and detect human impacts on ecosystems. Amidst rapid advancements, there is a pressing need for community-driven efforts to prioritize developments shaping the future of bioacoustics. Using a horizon scan approach, WILDLABS engaged 50 global experts in bioacoustics, representing diverse geographic, taxonomic, and industry perspectives, to identify and prioritize innovations likely to influence the field over the next two decades. Through an iterative process, 20 emerging priorities were identified and categorized into three key domains: engineering (e.g., affordable recorders, real-time data transfer, sound localization), analytics (e.g., individual identification, robust machine learning models, abundance estimation), and coordination (e.g., data-sharing frameworks, capacity building, integration into decision-making). These priorities highlight the growing potential of bioacoustics to address global challenges such as biodiversity loss and climate change, provided that barriers to adoption are addressed. Advancing the field will require strategic investments, interdisciplinary collaboration, and the development of sustainable solutions for data storage and accessibility. Furthermore, mainstreaming bioacoustics into environmental policy and decision-making will require the establishment of standardized protocols and equitable global capacity building. This roadmap underscores bioacoustics' transformative potential for biodiversity conservation and offers a unified vision to direct innovation, research, and global adoption of these technologies.

P1-47

Choosing pretrained bioacoustic models for downstream tasks

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The number of available pretrained bioacoustic models is steadily increasing, enabling researchers to incorporate these into their own pipelines and target datasets. A growing practice is to split pretrained models into feature extractors and classifiers, and use the trained extractor as backbone for a range of downstream tasks, more or less related to the training domain. However, current methods to identify the best model for a specific target domain remain limited, particularly in scenarios with large domain shifts. In this work we propose a unified evaluation process that assesses the feature extractors independently of their classifiers. Using qualitative evaluation of embedding spaces and quantitative evaluation of clustering and kNN classification, we have streamlined the process of evaluating and comparing the outputs of these models. This enables a more informed selection of the best supporting models for any given research question. Additionally, we present a python repository that standardises the feature extraction process for a multitude of pretrained models. It currently includes 15 state-of-the-art bioacoustic models, spanning a wide variety of model architectures, training paradigms and training domains. Researchers can obtain embeddings across all these models and investigate which, trained on insect sounds, bird sounds or general audio, is the most suitable for their specific questions. We believe this approach is a step towards a fair comparison of pretrained bioacoustic models in regard to adaptability and generalization, independent of the specific classes a model was trained on.

P1-48

Humans are animals and speech is bioacoustics: a computational perspective

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Bioacoustic research is adopting deep learning methods requiring ever-increasing amounts of data. In this context, we advocate for the use of speech — the most studied form of animal vocalization — as an asset for future computational solutions. While human vocal communication differs from other animal sounds, recent studies have demonstrated its potential in developing bioacoustic foundation models. Speech processing benefits from decades of research, yielding high-performance models, annotation frameworks, and self-supervised tasks transferable to bioacoustics. Notably, self-supervised transfer learning from speech has proven successful, while new model architectures and pre-training setups are continuously being published. Speech enables sequential encoding of information due to its inherent contextual nature and the architectures designed to tackle it. Although anthropocentric bias is an important concern, speech models do not necessarily rely on verbal content nor phoneme or word-level segmentation, mitigating some of its effects. Additionally, speech datasets often contain extensive recordings of isolated individuals, providing large clean focal data compared to passive acoustic monitoring. Speech-based models can thus be significantly larger and effective in focal settings, though they may struggle with chorus-like overlapping vocalizations.
We provide an overview of speech datasets and models characteristics as well as their potential usage in bioacoustics. We also present results from several studies highlighting the advantages and limitations of the approach in light of different acoustic scenarios, tasks and taxa. Finally, we suggest that further understanding of transfer from speech and language processing techniques to bioacoustics may offer new insights into the structure and evolution of communication across species.

P1-49

Investigating the structure of the greylag goose vocal repertoire: what can unsupervised methods tell us?

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Defining a comprehensive signal repertoire is the first step to understanding a vocal communication system. Here, we set out to investigate the vocal repertoire of one of the best-investigated model systems in ethology: the greylag goose (Anser anser). We used a large dataset of vocalisations collected over the last four years from a free-living population of greylag geese to investigate the acoustic structure of this species' vocal signals. To this end, we extracted four different types of data representations, which were projected into two dimensions using a frequently applied algorithm and clustered using two commonly used methods. In addition, we applied a graph-based clustering approach — Leiden community detection — which, to our knowledge, has not previously been used in bioacoustics. Our analyses revealed a partly graded vocal repertoire broadly matching early descriptions of the calls present in our dataset. Audio feature vectors revealed clusters most congruent with human labels and offered the most detailed visualisation of the acoustic space. Leiden performed comparably to established approaches but matched the number of human-defined classes closest, with less variable results. These findings highlight the impact that data representation can have and provide a quantitative characterisation of the greylag goose vocal repertoire.

P1-50

AI-assisted bioacoustics: from specialized models to multi-task cross-taxa LLMs

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To bridge this gap, a suite of methods aimed at enhancing model robustness and generalizability is explored. These include data augmentation techniques tailored to the audio domain, pseudo-labeling strategies that leverage weakly or unannotated data, knowledge distillation to transfer insights from larger models or ensembles, and neural network architectural adaptations specifically designed for acoustic pattern recognition.

P1-51

Geckording: Individual and spatial acoustic signatures in the advertisement calls of the red-spotted tokay gecko

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Prior research has indicated that the advertisement calls of the Tokay Gecko (Gekko gecko), used for mate attraction and territorial defense, differ structurally among distinct populations of Red-spotted and Black-spotted Tokay Geckos, likely due to genetic differentiation, environmental pressures, and biogeographical factors. Pronounced intraspecific genetic structuring has also been documented in Red-spotted Tokay Geckos in parts of mainland Southeast Asia, influenced by geographic origin, spatial separation, and natural barriers. It can be expected that ecological and biogeographical contexts also contribute to acoustic variation. Despite this, data on advertisement call variation remain scarce across much of Southeast Asia, particularly in the Indonesian and Philippine archipelagos, which represent natural geographic barriers. To address this gap, we launched the Geckording project, which combines original field recordings with citizen-contributed data, primarily from the iNaturalist platform and collected by both motivated and independent observers. The project aims to investigate individual and spatial variation in advertisement calls across a broad biogeographical gradient. Preliminary findings suggest both individual and regional differences in vocalizations and indicate biologically relevant variation that may reflect underlying genetic and ecological differentiation. However, the inconsistent quality of opportunistic recordings presents methodological limitations, emphasizing the need for more standardized approaches to acoustic data collection. Nevertheless, given the ongoing population decline driven by overexploitation for traditional medicine and the pet trade, as well as habitat loss, and considering the potential genetic consequences of further fragmentation, these initial results—despite the methodological constraints—provide a valuable foundation for future conservation efforts.

P1-52

Hullabaloo at the Zoo: Aligning acoustic research with the goals of the conservation zoo

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The role of the zoo has changed substantially since animals were first kept in captivity; modern conservation zoos are now international centres for the conservation of biodiversity. In 2023, a new, updated model for conservation zoos was proposed, with goals encompassing the wide-ranging impacts of zoos at local, national, and international levels including 1) Species and habitat conservation, 2) Education and training, 3) Scientific research, 4) Public health and wellbeing, 5) Conservation finance, 6) Economy and 7) Policy. A wide spectrum of acoustic research is applicable in the modern zoo, including noise and soundscape analysis, production and perception of sound by animals, behavioural aspects of vocal communication, as well as technical aspects such as sound localization, detection and classification. There is increasing interest among zoo conservationists in the use of acoustic research in animal husbandry and management, conservation breeding programmes, visitor experience, and education. Despite this interest, acoustic research in zoological settings has lagged behind that in other areas. Zoos often lack the capacity or funding to conduct large-scale acoustic research. However, there are many opportunities for acoustic researchers to align their research with the needs and wants of the zoo community, and to explore potential new areas of research with real-world impact on species conservation. Here, we explore the seven goals of the conservation zoo to show where collaboration could be beneficial for both zoos and researchers, and highlight the most productive ways to engage with zoo-based research and the wider zoo community.

P1-53

Estimating call densities of Bryde's whales in the Marianas region using the "Biotwang"

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Ocean bottom seismometers (OBSs), which are typically used to detect earthquakes, can instead be used as an opportunistic sensor to detect powerful, low frequency whale calls. The "Biotwang" call, which was recently attributed to Bryde's whales in the Marianas region of the Pacific Ocean, is of low enough frequency to be detected by OBSs. In this study, we are using data from a network of OBSs deployed from 2012-2013 near the Marianas Trench to investigate the seasonality of Bryde's whale calls and calculate their spatial densities using distance sampling. We have visually confirmed occurrences of this "Biotwang" call within the data. Using metrics from these known calls, we will make a synthetic template call and utilize spectrogram cross correlation to automatically detect "Biotwangs" recorded on the OBSs. The distance to each call will then be calculated using multipath ranging. Multipath ranging was developed to estimate distances to fin whales using calls recorded on a single OBS. Distances are estimated through measuring the recorded timings of call echoes (multipaths) from seafloor and sea surface reflections and comparing them with predictions from a sound propagation model. We have observed multipaths of "Biotwangs" on Marianas OBSs which suggest multipath ranging would also work for this species. We will then apply point-transect distance sampling to ranges to estimate seasonal changes in "Biotwang" spatial call densities. This showcases the utility of density estimation through multipath ranging and highlights broader possibilities in using OBSs as a passive acoustic monitoring tool in for characterizing whale populations.

P1-54

Evaluating eight years of the African bioacoustics community

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Established in 2018, the African Bioacoustics Community (hereafter ABC) was created as a forum for networking, collaboration and capacity building across the field of bioacoustic science. Principal activities include a biennial conference, freely available training materials and workshops, as well as science communication and public awareness activities. A bimonthly listserv established in 2023 has grown in popularity and now shares to over 500 registered participants. Recent additions to the portfolio of outputs include a verified sound library of African species, several large-format public art pieces to encourage mindful listening in nature and a creative multimedia workshop to promote engagement with natural sound. Through partnering with the Organisation for Tropical Studies (OTS), the ABC has developed and implemented a 14 day residential field course teaching applied bioacoustics in theory and practice, with plans for university accreditation in future iterations. To date, the majority of work has been undertaken by a core group of postgraduate students and early career scientists, supported through a modest stream of grants and awards, as well as conference generated revenue. One year of seed funding provided by the Richard Lounsbury Foundation was instrumental in formulating key roles, endeavours and outputs. As we look forward to the 5th ABC conference to take place in 2026, this talk highlights the significant achievements of a small, dedicated group of individuals with complementary skill sets and a shared vision.

P1-55

Temporal partitioning of bladder grasshopper calling activity to avoid predation

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Ecological interactions, including competition, predation, and environmental conditions, may significantly impact acoustic signalling behaviour. Here we characterise nocturnal signalling patterns in the bladder grasshopper Bullacris unicolor and relate this to biotic and abiotic factors, thus providing insights into ecological drivers of acoustic signalling behaviour. Passive acoustic recorders monitored nocturnal calling activity of the focal species, competitors and predatory bats across the distribution of B. unicolor. Results indicate that B. unicolor calls preferentially at certain times of the night, but that peak activity period varies across the geographic range. There was a strong relationship between B. unicolor activity and bat activity. Bullacris unicolor populations further north showed an overall avoidance of bat echolocation call period, whereas further south an overlapping of call periods between B. unicolor and bat echolocation was observed. Bats at northern sites showed a distinct activity period early in the night and B. unicolor may thus reduce predation risk by shifting signalling activity to later in the night. In contrast, bats were active throughout the night at southern sites and B. unicolor did not delay calling activity in this region of its distribution, although the species did shift signalling slightly to reduce peak overlap with co-existing bladder grasshopper species.

P1-56

Songbirds show the Lombard effect in the wild

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Urbanisation is rapidly increasing, leading to higher levels of anthropogenic noise that can negatively impact the fitness of wildlife, particularly through the masking of acoustic communication. Acoustic communication is vital for survival, and birds living in noisy environments must adapt to ensure their vocalisations are heard. The Lombard effect is an involuntary noise-dependent regulation of vocal amplitude in birds and mammals. The Lombard effect strictly refers to changes in vocal amplitude, but is often accompanied by other vocal changes such as frequency increases. In this study, blue tits (Cyanistes caeruleus) living in urban parks, in proximity to high levels of noise were experimentally exposed to filtered white noise 0–3.5 kHz during song production. Amplitude and frequency from calibrated recordings were measured before, during, and after the noise playback. Our results show that blue tits increased both the amplitude and frequency of their songs in response to white noise exposure (The Lombard effect). Amplitude showed a strong positive correlation with noise level, while peak frequency demonstrated a weaker correlation. Interestingly, we also observed a significant decrease in both amplitude and peak frequency following noise exposure, suggesting an overcompensation in vocal adjustments. This is the first study to calculate the Lombard effect within individuals of a wild population.

P1-57

A generalized behavioural response threshold to sound exposure in harbour porpoises expressed as an audiogram-weighted sound pressure level

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Loud underwater sound sources, such as sonars, seismic airguns, and percussive pile driving, are known to disturb marine mammals tens of kilometres from the source. This causes a justified concern for the impact of such sources, in turn creating a need for assessment of the likely magnitude of the impacted area prior to permitting. Impact ranges depend not only on the source itself (source level, frequency content etc.) but also on environmental factors affecting the sound propagation (bathymetry, sediment properties, hydrography etc.). The best way to include these factors in assessment is by combining acoustical modelling of received sound levels with generic response thresholds for different groups of marine mammals. Deriving such generic thresholds has proven difficult. Here, however, a generic threshold for behavioural reaction to noise (negative phonotaxis) is presented for the harbour porpoise, derived by compiling results from many empirical studies where onset of response could be coupled to received levels of noise. Across different sources, the responses were observed at received sound pressure levels (Lp,rms 125ms) weighted with an auditory frequency weighting curve resembling an inverted porpoise audiogram, in the range from 95 to 115 dB re 1 µPa. This threshold region corresponds to a loudness level between 50 and 70 dB Son (dB above the hearing threshold). Work funded by the Danish Energy Agency.

P1-58

Seasonal changes to the broadband underwater soundscape in a high-Arctic fjord, Inglefield bredning, Northwest Greenland.

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The Arctic is undergoing rapid environmental transformation as sea ice retreats due to climate change, fundamentally altering both marine ecosystems and potentially their soundscapes. As longer open-water seasons allow increased human activity and expose the ocean to more wind, waves, and calving icebergs, both anthropogenic and natural noise levels are hypothesized to increase. This study tests how ambient noise levels vary between ice-covered and ice-free conditions in a high-Arctic fjord, and what this means for future high-Arctic soundscapes in a warming climate. Noise loggers (SoundTrap ST600, 196 kHz sampling rate, duty cycle of 20 min/hour) were deployed over two years from eight moorings in Inglefield Bredning, Northwest Greenland. We show that mean ambient noise levels increase by 20–30 dB during ice-free periods in the 100-1000 Hz and 1-10 kHz frequency bands, alongside substantial spatial variability likely shaped by proximity to both glacier fronts and areas of human activity such as nearby settlements. Establishing these seasonal and spatial patterns in the Arctic acoustic baseline reveals how the presence or absence of sea ice shapes underwater noise levels, offering crucial insight into the timing, intensity, and variability of acoustic conditions that ice-adapted marine species must navigate. This study offers a foundation for future assessments of anthropogenic influence in a rapidly changing Arctic.

P1-59

Porpoises may be masked by recreational vessels, but they are unlikely to experience temporary threshold shifts (TTS)

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Porpoises are exposed to vessel noise in coastal habitats and may therefore be vulnerable to masking, especially from recreational craft which generate copious amounts of high-frequency noise. Alongside masking effects, this exposure could cause TTS, which is a regulatory threshold for noise exposure in marine organisms. To investigate masking and TTS in harbor porpoises from high-frequency noise, we used click-evoked potential measurements on a porpoise during exposure to 1/3-octave noise at 125 kHz compared to periods before and after noise exposure. When noise exposure coincides with the test signals, 125 kHz 1/3-octave noise with a level of 85 dB re 1µPa (rms) starts masking the hearing of clicks and no evoked auditory potentials resulted when the noise level reached 120 dB re 1µPa (rms). Masking in the echolocation band of porpoises may thus occur up to several hundred meters from some vessels. After noise exposure levels of 148 dB re. 1µPa2·s at 125 kHz, neural responses to the click stimuli were unchanged from baseline levels, rendering it unlikely that high-frequency components of vessel noise can cause TTS in a porpoise, even close to passing vessels as exemplified by our actual recordings of such. This could be explained by our finding that porpoises actively reduce their hearing sensitivity in anticipation of noise exposures. I conclusion: masking from vessels is indeed an issue in heavy traffic, but TTS is unlikely to be caused by exposure to vessel noise in the wild.

P1-60

Effects of vessel noise on ringed seal (*Pusa hispida*) acoustic communication

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The decline in sea ice concentration driven by climate change has been associated with a growth in vessel traffic across the Arctic. This has raised concerns about the vulnerability of Arctic marine mammals, which once inhabited quieter environments, to rising levels of underwater noise from increased vessel traffic. High levels of noise are considered a threat to marine mammal populations all over the world, potentially masking critical biological signals and disrupting communication. This study seeks to examine the effects of vessel noise on the acoustic communication of ringed seals in the Arctic through playback experiments conducted near the community of Ulukhaktok, NT, Canada, in May 2024 and May 2025. The experiments involved deploying SoundTrap acoustic recorders and automated underwater speaker systems in different leads in the sea ice where ringed seals were observed to haul out. Ten-minute exposure sessions alternated between underwater broadcasts of vessel noise treatments (chosen from pre-recorded sounds of three different vessels) and control treatments (pre-recorded natural ice sounds). All playback sessions were conducted with a minimum interval of 4 hours, occurring at 12:00 PM, 4:00 PM, 8:00 PM, 12:00 AM, 4:00 AM, and 8:00 AM. The analysis will compare ringed seal vocal behaviour before and after exposure sessions, focusing on changes in calling rates and call acoustic parameters to assess potential strategies used to cope with vessel noise and minimize masking. These findings will enhance our understanding of how increasing underwater noise levels from vessel traffic may affect the acoustic communication of ringed seals.

P1-61

Fright in Flight: Startle response in wild free-flying bats, following the onset of noise and light

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The level of anthropogenic pollution, such as man-made light and noise sources, continues to increase in natural habitats and could critically alter the way animals sense and interact with their environment. Most bats rely primarily on echolocation and secondarily on night vision to forage and navigate in low light habitats, making them especially vulnerable to sensory pollution from anthropogenic sources. Bats' retinae are rod-dominated, making them particularly susceptible to artificial light at night which could negate their ability to use their vision for navigation, whilst simultaneously increasing the risk of visual predation. Captive bats employ a Lombard response to the onset of noise, however in natural settings wild bats have the option to avoid noise altogether. Information on how wild bats react to anthropogenic stimuli in nature is still scarce, making it difficult to quantify how humans affect them. In this study we use 3D tracking cameras to assess how wild free flying Myotis spp. and Pipistrellus spp. react spatially and temporally to the onset of white light and broadband noise in their natural habitat. This high-resolution data allows us to quantify genus-specific startle responses in wild bats and create a dose-dependent curve, shedding light on the consequences of different received levels of light and noise on bats. Our results highlight that both Myotis spp. and Pipistrellus spp. respond negatively to the onset of anthropogenic light and noise, ultimately quantifying the magnitude of disturbance required to perturb wild free-flying bats in their pristine, quiet, low-light environments.

P1-62

Dynamic modelling of active sensing systems

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Active sensing animals gather information about their surroundings by emitting probes of energy and analysing the modulation of these probes by the environment. When multiple active sensing individuals are in close proximity, the emitted energy pulses can interfere with each other, making it more difficult for an individual to detect objects around them. This phenomenon, known as jamming, presents a significant challenge to collective behaviour in bats. Sound emitted by bats generates multiple echoes from surrounding objects and the bat needs to segregate these auditory sources into one coherent spatial scene. This problem is amplified when flying in groups: an individual needs to distinguish between the echoes of its own calls and the echoes of its neighbours' calls. Despite this, many bat species are social and often fly in large groups. So, how do bats manage to fly without colliding, even while experiencing jamming? To answer this question, we develop a theoretical model of biologically parameterized groups of active sensing animals. Building on static simulations of Beleyur & Goerlitz 2019, we create a dynamic model to understand how jamming affects individual flight and the overall collective flight behaviour. We explore the parameter space to find regions that minimize the cost variables (e.g. energetic consumption, collision rate, jamming) and compare it to currently available experimental data. By studying the sections of the parameter space realized in biological systems we identify general principles guiding active sensing in groups.

P1-63

RO-BAT: A bat-inspired robotic platform for collective auditory navigation and dynamic swarm sensing

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The Ro-BAT is a bio-inspired robotic system engineered to navigate its surroundings using solely auditory cues. It depends exclusively on a microphone array to gauge both the direction of arrival (DOA) and the intensity of sounds—whether passively received from other agents' calls or actively through obstacle echoes. Constructed from inexpensive, off-the-shelf electronic components and integrated with sophisticated DOA algorithms, this platform is designed to form a fully parameterizable swarm of bat-like robots. This setup enables a controlled exploration into collective sensing, offering the opportunity to continuously monitor and modify bat calls, thereby testing and implementing autonomous decision-making within a swarm and investigating individual behaviour in the so-called 'cocktail party nightmare'. The project is focused on two concurrent paths: firstly, the passive Ro-BAT, which was initially developed and validated in the audible range using a single functional robotic unit, has now been scaled to multiple units to study real-time passive localization and avoidance strategies in both bats and active sensing agents. Secondly, the active Ro-BAT, operating in the ultrasonic spectrum, is capable of emitting and receiving ultrasound calls to closely emulate the echolocation of various bat species. With its naturally tuneable and fully parameterizable design, this platform can adjust call intensity, duration, and rate, while also processing echoes from multiple objects around it. Both versions of the Ro-BAT have already proven their ability to locate and avoid external obstacles while effectively handling real-time input in complex environments.

P1-65

Temporal structures of calls and songs can tell us something about the caller's identity - and more!

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Avian communication often consists of complex sequences of sounds. The temporal patterns of these sequences can be a very powerful dynamic channel for encoding important social information, given that they may be less constrained by anatomy and less susceptible to partial signal filtering than e.g. frequency parameters. But do temporal patterns carry information about caller identity? Here, we explore how two species with distinct ecological contexts use temporal features in their vocal displays. The Little Auk (Alle alle), a pelagic seabird, uses complex calls in long distance communication, while the Yellowhammer (Emberiza citrinella), a farmland passerine, sings its songs in short-range territorial displays. We show that the temporal patterns of both species' vocal displays carry important social information, such as the caller's identity, and can do so without taking spectral parameters into account. Additionally, we show that these individual differences may occur not globally (e.g. within a population), but rather at scales behaviourally relevant to the animals (e.g. across the neighbours). We suggest that by integrating detailed temporal analyses into studies of information in animal signals, we can refine our understanding of animal communication networks and their social complexities.

P1-66

Non-siblings' individual vocal discrimination of group members in Psittacus erithacus

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Recognising individual partners is a crucial survival challenge linked to cooperative and prosocial behaviours. Such behaviours have been observed predominantly in animals with complex cognition and social systems, such as primates and dolphins. Among birds, parrots stand out for their complex cognition and propensity for cooperativeness. Some indications suggest that Psittaciformes can differentiate between their conspecifics. This study investigates Psittacus erithacus, a species known for its cognitive abilities, to determine its capability of recognising conspecifics using vocal cues alone. We employed a playback experiment using the habituation/dishabituation paradigm, comparing individuals with neutral social affinity. Our findings indicate that the subjects could effectively discriminate non-kin group members even in the absence of strong social closeness.

P1-67

Post-buzz whistles may signal attributes of prey within foraging groups of long-finned pilot whales

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As group-living animals, long-finned pilot whales (Globicephala melas) in the Alboran Sea and Strait of Gibraltar often synchronize foraging dives within their groups, allowing them to maintain acoustic contact at depth. However, it is unknown if they communicate during dives to mediate foraging decisions. We deployed sound and movement recording DTAGs in small groups of pilot whales to understand the context of social signaling during foraging. On 73 DTAGs deployed on pilot whales from 2010-2015, 3.9% of echolocation foraging buzzes (n=3160 total buzzes) were followed closely by a high-frequency whistle. These whistles were characterized by peak frequencies typically >7 KHz, in contrast to more common low-frequency calls typically below this threshold. We tested whether post-buzz whistles were associated with specific foraging contexts by modeling their presence after each buzz (putative prey capture attempt) and on each foraging dive. We detected 124 post-buzz whistles on 36 DTAG records and 95 deep (>200 m) foraging dives. We used a binomial GLMM to test whether the inter-buzz interval, buzz duration, and peak jerk during the buzz predicted the presence of post-buzz whistles. We also modeled whether maximum depth, dive duration, and buzz rate predicted post-buzz whistle presence during each dive. Whistle presence was significantly associated with a longer duration of the preceding buzz ($p<0.001$) and a higher buzz rate during the dive ($p<0.01$). We cannot rule out other functions, but the association of whistles with buzz- and dive-level covariates suggests they may communicate attributes of prey quality or availability to nearby group members.

P1-68

Investigating vocal patterns in Panthera species: rhythm and vocal exchanges in lion (*Panthera leo*) and tiger (*Panthera tigris*)

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This study aims to investigate whether lion (Panthera leo) and tiger (Panthera tigris) vocalizations exhibit rhythmic patterns. In fact, rhythm is a putative musical universal and can convey relevant information from an eco-evolutionary perspective. We analyzed 206 recordings from a group of seven individuals (five lions, two tigers). We marked sounding intervals, then inter-onset intervals were calculated from vocal onsets. We computed the rhythmic ratios (rk=tk/[tk+tk+1]) and verified whether the distribution of rk values exhibited peaks around 0.5, a value meaning that all intervals within a sequence are equally spaced. Our findings reveal that both species display a rhythmic pattern with a distinct isochronous component (1:1), even at individual level, suggesting enhanced signal efficiency and communicative function. To support this hypothesis, we also analyzed vocal exchanges between individuals. Firstly, we run Granger causality test to verify whether the onset timing of one individual's vocalizations predicted that of another. Instances of temporal influence were observed, also between individuals of the two different species. However, this feature emerges in a limited subset of vocal exchanges, suggesting that coordination does occur, but it is not a prevalent characteristic. Additionally, to investigate the transitions between vocalizing individuals, a permutation test was performed. Our results show that certain individual transitions occur more than others at a significant level. These findings support the role of rhythm in social communication dynamics. This work contributes to the broad investigation of rhythmic structures in mammalian vocalizations and the evolution of acoustic signals in closely related yet behaviorally distinct species.

P1-69

Bottlenose dolphins show clear geographic variation in whistles when controlling for repeated, stereotypical signals

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Geographic variation in animal signalling has been described in many taxa, including insects, anurans, birds and mammals. Delphinids exhibit large acoustic repertoires, but the presence and extent of their geographic variation remains largely unknown. We analysed geographic variation in whistles of the bottlenose dolphin Tursiops truncatus, utilising 46 time-frequency measurements from 980 whistles across 7 sites within the Atlantic and Pacific Oceans. Bottlenose dolphins are known to produce individually specific signature whistles which are highly stereotyped. Here we control for the presence of signature whistles by excluding whistle type repetitions to avoid potential bias towards inter-individual, as opposed to inter-site, differences. We employed Random Forest classification modelling to establish a) whether whistles are reliably classifiable to site of origin; b) whether inter-site variation exceeds intra-site variation; and c) whether this variation is present between two adjacent populations with some overlap in ranges. This inter vs. intra-site comparison is a vital requirement to delineate geographic, from individual, variation. Whistles were accurately classified to site of origin (71.6% ± 4.14 [1 SD], k = 0.66). Our intra-site model performed poorly (28.0% ± 3.68, k = 0.16), indicating that inter-site differences outweighed intra-site differences. Our final model accurately classified whistles to the adjacent Eastern and Western Scottish coasts (81.3% ± 6.52, k = 0.63). These results inform ongoing development of odontocete species classifiers, and vocal differences observed between adjacent populations in this study may indicate social delineations or Evolutionarily Significant Units (ESUs) that would require specialised consideration from policy makers.

P1-70

The whistle repertoire of west coast Scottish common dolphins (*Delphinus delphis*)

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Vocal repertoires are catalogues of the sound types a species produces, and their diversity can be an indicator of communication complexity. Delphinids use whistles for communication, but comprehensive whistle repertoires are unavailable for many species. Existing research has mostly focused on describing means and ranges of acoustic variables, or very broad categorisation of contour shapes, rather than an analysis of individual whistle types. However, we know from bottlenose dolphins that the frequency modulation pattern carries significant information. This project describes the detailed whistle type repertoire of the short-beaked common dolphin (Delphinus delphis) from the west coast of Scotland. Whistles were measured and traced using PAMGuard's Real-time Odonotocete Call Classification Algorithm (ROCCA) module and the resulting contours were analysed using ARTwarp. A total of 1271 whistles were extracted and categorised from 29 independent encounters, resulting in 76 whistle types containing two or more whistles. The cumulative whistle type discovery curve levelled off at these 76 types when 600 whistles were analysed, and therefore we are confident that the full repertoire has been described. We also used this whistle repertoire to assess the use of biodiversity indices to predict repertoire size from smaller whistle samples. We found that the most accurate prediction is achieved by using the species richness measure with a hill number of 0. By describing and comparing whistle types and whistle repertoires, we can obtain a more complete picture of delphinid communication and start to understand the factors that impact variation between and within species.

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P1-71

Geographic variation in false killer whale (*Pseudorca crassidens*) whistle characteristics and repertoire diversity.

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False killer whales (Pseudorca crassidens) are widespread across tropical, subtropical and temperate waters but are rare and poorly studied in most of their range. They produce whistles distinct from those of other species, but little is known about geographic variation of whistles within this species. This study explores geographic variation in the whistle characteristics and repertoires of false killer whales from three regions: Central America (CA, n=2915), Brazil (n=1177) and the Northwestern Hawaiian Islands (NWHI, n=1184). Within CA, microvariation was assessed across four locations: Costa Rica North (n=378), Costa Rica South (n=422), El Salvador (n=342) and Panama (n=581). Balanced random forest models were trained to classify whistles to encounter, location and region using seven time-frequency variables measured from spectrograms. Only 23% of whistles were correctly classified to 15 encounters within CA, 43% to 4 locations within CA and 57% to the 3 regions. In comparisons of whistle repertoires generated using artificial neural networks and dynamic time warping (ARTwarp), many whistle types were shared across locations and regions. Two-tailed permutation tests found fewer location-specific (meanpermuted=50.2%, $p<0.0001$) and region-specific (meanpermuted=50.1%, $p<0.0001$) whistle types than expected by chance. Biodiversity indices (species richness, Shannon and Simpson indices) were used to compare the diversity of whistle repertoires and showed no significant differences by location within CA. However, regional differences were significant, with CA showing the greatest repertoire diversity and NWHI the lowest. These findings show that the whistle repertoires of false killer whales are largely shared, even across distant geographic regions.

P1-72

The micro- and macrodialects of the parti-coloured Bat (*Vespertilio murinus*)

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Bats are highly social animals which use acoustic signals for prey detection, orientation, and social communication. Some bat species not only use simple calls in a social context but produce a complex song for mate attraction and/or territorial defence. This makes bats one of the few mammalian taxa that are known to sing. Unlike the well-studied song of songbirds, comparably little is known about bat song. Studies on regional differences (i.e.dialects) of bat song are sparse and often limited in scale. The parti-coloured bat, Vespertilio murinus, is a palearctic bat species that migrates long distances and also sings. Its song is mainly emitted during flight and highly characteristic for this species: each song motif is composed of two different syllable types, and motifs are repeated in long sequences. Here we present the first European-wide study on the song of the parti-coloured bat, conducted in 18 different countries. By comparing acoustic differences of song motifs from different locations in the same country and in different countries, we were able to study both small- scale differences (microdialects) and large-scale differences (macrodialects). In total, we analysed the acoustic parameters of more than 5000 song motifs. We found evidence for pronounced micro- and macrodialects in the song of parti-coloured bats. This study is an important foundation for future work on the functional significance of song dialects in a long-distance migrator.

P1-73

Using touchscreens for auditory preference experiments on zebra finches

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Birdsong is a complex and costly behavior, that (among other social functions) serves mate attraction in songbirds. Patterns of female birds' preferences to acoustic features in male song offer crucial insights into the evolution, ecology and physiology of birdsong and have been studied extensively in choice experiments, where e.g. two or more songs of varying properties are presented to a female bird. We aimed to increase the flexibility of previous paradigms by developing a Touchscreen Operant Conditioning Chamber to assess song preferences in female zebra finches. The focus was on obtaining the data autonomously, making the experimental paradigm simple to report and improve replicability. Our novel setup functioned well for groups of three birds and yielded data without inter- and intra-observer error. However, success rate of learning was low compared to previous studies, and interaction with the screen persistently decreased during the preference assay, creating false positives in the preference assessment. We think the low rate of success in our experiment could be inherent to touch screens, because pressing the screens surface was challenging for zebra finches. This means that simpler and previously established paradigms, such as string pulling or perch hopping, are more promising options for song preference studies.

P1-74

Marmoset vocalisations encode information about both internal states and external events

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It is often thought that animal vocalisations only convey information about either internal states or external events. In addition, it is debated whether communication about external events is in fact a simple by-product of the internal states that are triggered by these events. To experimentally address these issues, we analysed phee and ek vocalisations of common marmosets (Callithrix jacchus) produced in five treatments of negative, neutral or positive valence, and quantified the marmosets' arousal using nasal temperature change. We investigated the effects of the internal state (arousal and valence) and external events (food context and playback of conspecific vocalisation) on the acoustic structure of both phee (long- distance affiliative contact calls) and ek calls (mobbing/agonistic calls). Overall, phee parameters were affected by all factors, but mostly varied depending on whether the context was food-related or not. By contrast, ek parameters were only affected by arousal. Support Vector Machines (SVM) confirmed the findings for phees, successfully classifying these calls based on both internal states (valence and arousal) and external events (food context and conspecific calls), and revealed similar results for eks, with an accuracy exceeding chance levels. We thus find that information about both internal states and external events are simultaneously encoded in marmoset calls, similarly to human language. Most strikingly, marmosets encoded "food" in their phee calls independently of their own emotional arousal, and independently of whether they were allowed to consume food, were teased with food, or only heard playbacks of conspecific food calls without food being present.

P1-75

Rhythmic entrainment or note-by-note coordination: the adaptive mechanism behind extremely fast duetting in non-passerine species

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Spontaneous entrainment to rhythm or keeping the beat is the core of human musicality, and as such may have adaptive significance. Entrainment is also known from group behaviors of some invertebrates and amphibians, but it is controversial whether the benefits of synchrony achieved extend beyond individuals. Also, no convincing evidence has been provided for the use of entrainment in group signaling in other higher vertebrates, where coordination is the result of training and is based on a local call-and-response mechanism. Here we investigated the structure, mechanism, and function of duets in the little crake (Zapornia parva), a species of nonpassierine bird in which the mated male and female produce extremely fast alternating trills, where 23-ms pulses are repeated at ca. 30 Hz. Analysis of natural trills showed their high isochrony and perfect alternation in duets. When we played solo trills to unpaired birds, they joined them spontaneously to form duets despite the lack of any training. However, they failed to join non-isochronous solos with randomized interval lengths, suggesting that entrainment of intrinsic rhythms rather than note-by-note coordination is responsible for the formation of little crakes' duets. Finally, when in the experiment we played regularly alternating and irregular duets to territorial mated pairs, the former elicited stronger reactions. This indicates the importance of the alternation obtained through entrainment for territorial interactions. Overall, our results show that mated little crakes not only synchronize their extremely fast trills based on entrainment, but also that their cooperative behavior is adaptive.

P1-76

The battle and the serenade: Evaluating methods to validate birdsong's dual functions

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Birdsong is one of the most prominent sounds in nature, with its functions long studied under the "dual function" theory, which posits that male song serves to repel male competitors and attract females. This study reviews the definitions of bird vocalizations (songs and calls), early theoretical assumptions about song function, and methods for validating these functions across species. For repelling male competitors, four main methods were identified: direct observation, field playback experiments, mute experiments, and speaker replacement experiments. These methods provided evidence for 35 species across 21 families, with speaker replacement experiments offering the strongest interpretability by isolating the role of song in territorial defense. For attracting females, six methods were analyzed: direct observation, field playback experiments, phonotaxis experiments, copulation solicitation display assays, parental behavior assays, and mute experiments. Evidence was found for 42 species in 23 families, with phonotaxis experiments (in field settings), copulation displays, and parental behavior assays providing the most direct evidence due to their close linkage to reproductive outcomes. Our analysis highlights gaps in taxonomic coverage, with only 15% of passerine families represented in current studies. Despite the dominance of the "dual function" theory, the limited taxonomic range and reliance on indirect evidence underscore the need for broader species sampling and more robust experimental designs.

P1-77

Project ELECOM – Decoding elephant communication with AI

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Wild elephants live in fluid fission-fusion societies, often spacing themselves hundreds of meters, sometimes kilometers apart. Acoustic communication is therefore crucial, and their pronounced vocal plasticity lets them adjust call structure to shifting environmental and biological contexts. This vocal flexibility creates significant within-call-type diversity, which makes the interpretation of biologically meaningful cues from their vocalizations particularly difficult. The ELECOM project (Decoding elephant communication using AI) joins biologists, acousticians, mathematicians, and computer scientists to tackle this challenge. By leveraging novel machine learning methods and one of the world's largest annotated acoustic datasets of captive and free-ranging African savannah elephants (Loxodonta africana), we aim to automatically detect, classify, and identify acoustic patterns within a combined data- and knowledge-driven framework. Using auditory brainstem response (ABR) and other hearing tests we will improve the current knowledge of the species' full auditory range to generate precisely calibrated, time–frequency representations specific to elephants, ensuring accurate extraction of biologically relevant acoustic information. These calibrated representations will directly inform our analysis and guide the synthesis of realistic calls, which will subsequently be tested in controlled playback experiments to verify how context-related information is encoded in elephant vocalizations. The project's ultimate aim is to create a first "dictionary" of elephant vocalizations, comprising identified and verified acoustic features along with their potential meaning. The ELECOM project will offer novel methods and tools for behavioral ecology, conservation monitoring, and transferable frameworks for decoding other complex animal vocal communication patterns.

P1-78

Eupsittula parakeets' ability to modify response calls

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Eupsittula parakeets live in fission-fusion flock dynamics. Vocally interacting with individuals that they are more or less familiar with is crucial for them to find resources and obtain a position in their flock. They produce individually distinctive baseline calls (loud contact calls), but when interacting vocally during fission and fusion events, they can alter the structure of their contact call to imitate others. Previous research has shown that they can adapt their response calls in different directions. Based on these findings, we investigated the parakeet's capability to modify their calls in response to a changing call from an unknown conspecific. To test the individual's ability to modify their own call to match a changing stimulus, 5 peach-fronted conures E. aurea was given each 6 playback tests investigating the bioacoustics parameters: fundamental frequency, formants and duration. For each parameter 2 test were given to both investigate the lower range of the parakeet's spectrum and for the higher range. These results reveal some of the complexity of these species and their communication systems.

P2-1

The silence of the lake: The underwater soundscape of the endangered Saimaa ringed Seal

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Soundscapes provide valuable insights into ecosystem dynamics and how changes to the environment can shape ambient noise levels. Studying these changes is especially important for endangered populations, particularly those relying on sound for communication, navigation and foraging. The endangered Saimaa ringed seal (Pusa hispida saimensis) utilises sounds for communication during the breeding season, including mother-pup interaction, mate attraction and territorial defence. This critical period occurs during the ice cover period in early spring, providing a relatively undisturbed acoustic environment to facilitate communication. However, global climate change threatens to alter this soundscape through reduced ice cover, earlier ice breakup, and increased human access to the lake. In this study, we used passive acoustic monitoring to measure ambient noise levels and characterise sound sources at four breeding locations within Lake Saimaa. We provide baseline data on the soundscape during the Saimaa ringed seal breeding period. Preliminary results suggest that the main contributors to the soundscape are weather, with little influence from biological or human activities. In general, the environment can be considered relatively pristine. This study serves as a baseline for future studies to understand the impact that changes to climate and human activities could have on the environment and behaviour of the Saimaa ringed seal.

P2-2

Bird-generated soundscape as an indicator of environmental quality and its changes

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Over the past 20 years, Europe has experienced significant changes in bird abundance, distribution, and species composition. These transformations vary geographically and across habitats, shaping interspecies competition for resources, including acoustic space. Bird vocalizations play a crucial role in communication, and the emergence of new species may disrupt acoustic stability, potentially serving as an indicator of ecological change. In this study, we analyzed changes in the use of acoustic space by birds in Poland from 2000 to 2023. Data from the Common Breeding Bird Monitoring program were used to assess species composition on randomly selected transects. Based on recordings from Xeno-canto, the acoustic profile of each of the 214 bird species was determined, and an Acoustic Space Evenness Index was proposed to describe the use of acoustic space by entire bird communities at each transect. The index values indicate the degree to which different bird species occupy specific frequency bands. Generalized Linear Mixed Models were applied to examine how the use of acoustic space by birds has changed over time, across habitat types, and in relation to protection status. The results show that the use of acoustic space by birds has changed over the years. Higher acoustic evenness was observed in farmlands compared to woodlands, suggesting that farmland ecosystems have undergone more rapid changes in recent decades. Additionally, the findings indicate that protection status has no effect on acoustic evenness. This approach provides insight into how avian communities shape the acoustic environment and how these structures evolve in response to ecological change.

P2-3

From brain connectomes to soundscape connectomes: A graph-based unsupervised approach for acoustic heterogeneity identification

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Graph-based approaches have proven invaluable for uncovering complex relationships across scientific fields, notably in neuroscience for constructing brain connectomes, comprehensive maps of neural pathways that enhance our understanding of cognition and disease. Inspired by these methods, we propose an unsupervised framework to build "soundscape connectomes," capturing relationships, similarities, and differences across diverse geographic sites by leveraging biophonic information. Our approach decomposes audio recordings from passive acoustic monitoring (PAM) into sonotypes, unique acoustic entities characterized by specific time-frequency features. These sonotypes form a site-specific "acoustic footprint," which feeds into graph inference algorithms, where nodes represent locations and links encode acoustic similarities. Since our goal is to uncover non-obvious links among locations, the absence of labeled data presents challenges for reliably training state-of-the-art graph models. To address this, we employ flexible initialization strategies, propose an unsupervised evaluation metric for model selection, and implement stability-focused mechanisms to preserve consistent core links. This framework was tested on PAM datasets collected from Colombian regions, each featuring varying vegetation covers. The resulting soundscape connectomes reveal ecological patterns consistent with expected land cover types and expose relationships that may remain overlooked by standard remote-sensing techniques. This advantage arises from acoustic signals' ability to capture subtle, understory-level changes, such as shifts in the presence or vocal activity of biophonic sonotypes, providing an added layer of ecological insight. Our proposal offers a promising avenue for identifying nuanced biodiversity patterns and guiding data-driven conservation strategies by extending connectome principles from neuroscience to soundscapes.

P2-4

Real-time performance evaluation of a smart acoustic buoy for right whale detection

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Increasing pressures from human activities are threatening species worldwide. Developing reliable, efficient methods for detecting and reporting relevant information on species presence can be an important tool for mitigating negative anthropogenic impacts on animals. Planned and ongoing offshore wind construction sites overlap with critical habitats for endangered North Atlantic right whales (NARW) which will increase the likelihood of potentially deadly ship strikes. Therefore, efficiently detecting and communicating NARW presence is a key component to developing offshore infrastructure in a minimally invasive manner. In this work, we have developed a real-time neural network for NARW upcall detection capable of running onboard a Medusa smart buoy that demonstrates significantly improved performance compared to existing real-time algorithms. During Summer 2023 and 2024, we conducted field trials in which we simultaneously deployed four Medusa buoys along with a playback speaker to validate the real-time detection performance of the Medusa NARW detection system under field conditions. We found that the full system is capable of reliably detecting NARW upcalls in real-time, recovering approximately half of upcalls above 1 dB SNR and approximately 90% of upcalls above 8 dB SNR. Furthermore, the full array demonstrated stable clock synchronization of approximately +/- 0.25 ms across a 4-hour test enabling potential downstream localization approaches. We quantified the theoretical impact of pile driving on detection performance and found that pile driving events do not trigger false NARW detections and will only reduce the detection performance of the algorithm if NARW vocalizations directly overlap with piling signals.

P2-5

Establishing a baseline for Hawaiian hoary bat (*Lasiurus semotus*) acoustic activity in the Māhukona navigation and ecological complex

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Passive acoustic monitoring is an essential tool for assessing the distribution and habitat use of cryptical species, like the endemic Hawaiian hoary bat (Lasiurus semotus). With the goal to establish a baseline for bat activity at the onset of habitat restoration, we conducted an acoustic survey in the Māhukona Navigation and Ecological Complex, a newly acquired reserve located on the island of Hawaiʻi managed by the Hawaiʻi Land Trust. In January 2025, we used two SM4Bat recorders to sample ultrasonic soundscapes across 16 sites spaced at a minimum of 200 meters apart, cycling units between sites to maximize spatial coverage. Recordings were continuous from sunset to sunrise. We used a custom classifier based on BirdNET embeddings created to detect hoary bat vocalizations over recordings. We interpolated detections in space to determine areas of higher density. We further explore the relationship between habitat vegetation structure and bat acoustic activity which can provide insights for the ongoing restoration of the area. We present these results in addition to a comparative methods analysis – evaluating BirdNET and Kaleidoscope automated detection models against manual call verification. This comparison will help determine the accuracy and reliability of automated classifiers in detecting L. semotus vocalizations, which are known to differ from mainland bat species. The results provide critical insights into habitat use patterns and inform long-term monitoring and conservation strategies for Hawaiʻi's only native terrestrial mammal.

P2-6

VALKYRIE: Harmonizing large-scale passive acoustic monitoring data of harbor porpoises through a Europe-wide initiative

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Recent technological developments have substantially enhanced our ability to monitor and store data from underwater environments using passive acoustic monitoring (PAM). Autonomous long-duration recording devices such as the C-POD and the update, the F-POD (Chelonia, Ltd), have increased the accessibility of using PAM for studying marine mammals, particularly the harbor porpoise (Phocoena phocoena) – a key indicator species for temperate coastal and shelf waters. Additionally, the narrow-band high-frequency clicks produced by porpoises are well-suited for PAM, as they can be readily detected and classified by simple means. These advancement in data collection, storage and processing provide new opportunities for analyzing large-scale movement and behavioral patterns. However, a key challenge remains: ensuring the successful aggregation and harmonization of PAM data across projects and institutions. The EU Digital Twin Ocean (DTO) initiative aims to address this by collecting historical and ongoing PAM data on harbor porpoises across Europe, facilitating large-scale monitoring of porpoise populations. As part of the DTO, we have developed a user-friendly tool, VALKYRIE, to gather, organize, and harmonize porpoise detection data for inclusion into the European Tracking Network (ETN) database. Participating partners will upload data files and accompanying deployment metadata via an app interface that directly extracts detection data and bypass export formatting discrepancies, ensuring harmonization of data. With the participation of a variety of European organizations, we aim to compile large-scale data to build a comprehensive understanding of the movement and behavioral patterns of harbor porpoises in European waters, both over time and into the future.

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P2-7

Temporal patterns in Delphinid foraging in the mesophotic reef zone of the Red sea

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Despite the ecological significance of cetaceans in marine ecosystems, their presence and foraging activity remain largely undocumented in the Red Sea—particularly within mesophotic reef zones. Delphinids, a highly intelligent and behaviorally plastic group of toothed whales, exhibit regionally specific traits that underscore the need for localized monitoring and management. Understanding their ecological role is especially critical along the understudied western coastline of the Red Sea. In this study, we present the first passive acoustic recordings of foraging delphinids in the mesophotic zone of the central Red Sea. Using Passive Acoustic Monitoring (PAM), we documented distinct diel and seasonal patterns of acoustic activity. Three primary types of fish chorusing were identified during both winter and summer, potentially influencing the timing and behavior of delphinid foraging events, as well as the species composition present. Our findings highlight the value of PAM as a complementary tool to visual surveys for tracking delphinid presence and behavior in reef-associated habitats, and support its application in future conservation strategies for these keystone predators.

P2-8

A Comparison between traditional bird survey and AI powered species call detection suggests high reliability of passive acoustic monitoring in Xitou, Taiwan

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Passive acoustic monitoring (PAM) is widely applied to soundscapes ecology. New techniques extract more detailed information. However, the differences between PAM and traditional survey methods are less understood. We simultaneously collected point-count bird surveys and PAM data during the breeding season in Xitou, Taiwan, with 67% of the surveyed species being vocal. Two deep-learning models, SILIC (focused on local species) and BirdNET (trained on a global database), were used to analyze the recordings. Manual re-surveys of the PAM data were also conducted. We compared the relative abundance of each species across in-situ point count, manual inspection, and the two AI models. The difference between in-situ point counts and manual inspection was minimal (mean = 0.44%), and SILIC's detection was closer to point counts (mean = 2.22%) than BirdNET's (mean = 6.57%). The species composition of SILIC detection was closer to point count data than BirdNET detection. Both SILIC and BirdNET performed well on recordings from Xitou. The differences between deep-learning models and point count resulted from detecting non-vocal species. These findings suggest that well-trained AI models like SILIC can provide reliable information in a bird community dominated by vocal species.

P2-9

Listening beneath the waves: southern elephant seal drift dives reveal the southern ocean soundscape

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The soundscape defines the acoustic environment animals live in, shaping key aspects of their behavior and communication. Measuring the underwater soundscape is crucial for characterizing these environments, yet current moni-toring efforts often rely on stationary recorders with limited spatial coverage. The Southern Ocean remains poorly sampled acoustically due to its remoteness and harsh conditions. Here we explore the use of deep-diving southern ele-phant seals (Mirounga leonina) as mobile passive acoustic monitors of this en-vironment. To do so, we analyzed acoustic data collected by long term sound recording tags deployed on nine southern elephant seals, focusing specifically on drift dives—periods when the animals are nearly motionless and silent. Our third-octave band analysis of the audio extracted from these drift phases re-veals persistent low levels of ambient noise, with occasional distinct acoustic events from biological or environmental sources. These results demonstrate that elephant seals can function as natural, low-cost autonomous underwater vehicles (AUVs), significantly enhancing our ability to monitor vast underwater soundscapes in remote oceanic regions.

P2-10

Integrating acoustics in environmental monitoring schemes: Defining the spatial footprint of acoustic monitoring

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Integrated environmental monitoring schemes are emerging globally due to the urgent need to address biodiversity loss, climate change, and habitat degradation. Comprehensive and large-scale monitoring is essential to detect environmental changes, guide conservation actions, and evaluate management strategies. Monitoring techniques typically have well-defined spatial footprints, i.e. the area being surveyed (aerial photos, vegetation plots, camera traps, etc.). However, the footprint of acoustic sensors varies substantially with species-specific factors, environmental conditions including weather, local topography, sensor specifications, etc. For instance, many calls and songs can be detected 100-200m away in good conditions. However, some calls can be heard from distances over 1km away (large mammals and birds) while others can reach only tens or even only a few meters (amphibians, bats, rodents, insects,). Detection distances can also vary for a single species and can be as much as twice as large depending on site vegetation or actual weather. Even though a monitoring scheme might be targeted to survey specific habitat types, in fragmented landscapes, recorders often detect species from outside the local habitat, complicating ecological inference. To seamlessly integrate acoustic monitoring into larger ecological monitoring frameworks, a better understanding of acoustic footprints is essential. Here, we present the design of an acoustic monitoring scheme targeting this issue and preliminary results illustrating that arrays of synchronized recorders can be used to effectively filter out sounds coming from outside a well-defined survey area. This approach leads to more accurate associations between habitat features and the soniferous animals using them.

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P2-11

Spatio-temporal patterns of bird communities inferred from application of machine learning on regional passive acoustic monitoring data

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Passive Acoustic Monitoring (PAM) has become increasingly popular as a practical and cost-effective method to collect robust biodiversity data in ecological research projects. Advances in autonomous recording technology have opened avenues to scale-up acoustic data collection at larger spatial and temporal scales. However, the main bottleneck in PAM-based projects is the post-processing time to extract relevant information from the recordings, which have often been processed with manual review and annotations. In this study, we evaluate the potential and current limitations of BirdNET-Analyzer v2.4, a deep learning algorithm for bird vocalization recognition, as a tool to assess bird community composition through the automated analysis of large-scale acoustic data. We analyzed these soundscapes using both BirdNET and listening by expert local birders and compared results to traditional point count surveys to evaluate algorithm performance for developing species lists. We estimated species-specific minimum thresholds that enabled high confidence for 120 species across 104 recording locations at a regional scale. Our analysis demonstrates that BirdNET identifications reliably represent the bird community across diverse habitats, including some cryptic species. The higher temporal resolution of acoustic data provides further insights into habitat occupancy of short- and long-distance migrants and seasonal habitat usage of different species. BirdNET provides the opportunity to significantly expand bird research and monitoring efforts and increase the accuracy and precision of occupancy, abundance, and population trend estimates, improving our understanding of population status and dynamics and our ability to conserve vulnerable species.

P2-12

Modelling acoustic detectability using simulated data to improve BirdNET species abundance estimates

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Machine learning methods such as BirdNET have transformed passive acoustic monitoring (PAM) by enabling large-scale detection and classification of bird vocalizations. These detections can serve as proxies for species abundance through vocalization rate estimates, yet variations in acoustic detectability across sites can introduce significant biases. Previous studies have assessed acoustic detectability using playback experiments, rangefinders, extensive manual validation, and other methods that are difficult to scale. We address this challenge by leveraging simulated data to model the relationship between BirdNET confidence scores and signal-to-noise ratio (SNR) in an automated, species- and site-specific manner. Then, by analysing detections within the PAM data to estimate both vocalization and background noise amplitudes (and subsequently derive SNR), our method characterizes how noise conditions affect detection probabilities, thereby estimating acoustic detectability with minimal manual validation. Finally, we combine the estimated detectability with vocalization rates to predict species abundance. We demonstrate the utility of our approach on datasets with co-located recorders and in-person surveys, providing a scalable framework for improving abundance estimates in large-scale PAM studies.

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P2-13

Automated acoustic individual identification of birds by vocal signature: a machine learning method and an application to Ovenbirds

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Individual-level recapture data enables the analyses of annual survival and abundance—two key parameters for characterizing demographic trends and assessing habitat quality. Individuals of many songbird species can recognize the vocal signatures of other individuals, suggesting passive acoustic monitoring (PAM) could generate non-invasive, scalable individual "capture" data. Recent machine learning advances show promise for distinguishing individuals from a known set using acoustic features. However, automated recognition of unfamiliar individuals in PAM data has not been accomplished. We present a method for automated individual identification of birds by acoustic signature. Using Ovenbird (Seiurus aurocapilla) as a model species, we first hand-labeled songs to individuals, and corroborated these labels by their alignment with spatial territories. We evaluate the accuracy of our automated individual identification approach, then apply the model to a four-year PAM dataset from Pennsylvania to assess annual apparent survival with a Cormack-Jolly-Seber model. Our method accurately assigns Ovenbird songs to human-labeled individuals (92%). Across our 4-year PAM dataset, annual song-based apparent survival is 0.68, suggesting that adult male Ovenbirds have relatively high year-to-year survival and site fidelity. Our approach to automated acoustic individual identification can be applied to various species, conditional on vocal learning characteristics and repertoire size. We share open-source code and data to support further use and development. We expect acoustic individual identification methods will enhance our ability to evaluate bird population demographics, supporting proactive management and conservation planning.

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P2-14

Ecoacoustic monitoring of avian communities in the Fukushima nuclear exclusion Zone: Establishing ecological baselines before large-scale environmental remediation

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Land abandonment represents a significant global change driver with complex ecological consequences. The unplanned human depopulation of agricultural and residential areas in Fukushima Prefecture following the 2011 nuclear disaster offers unique insights into ecological succession. As large-scale remediation efforts commence to support potential human resettlement, a brief window exists to document current ecological conditions of abandoned land within the exclusion zone. We deployed AudioMoth acoustic recorders within woodland, scrub-shrub, wetland, and abandoned agricultural habitats to establish comprehensive avian biodiversity baselines before remediation-induced disturbance by demolition and clean-up crews. Using BirdNET and NatureLM-audio models to classify species' vocalizations and characterize bioacoustic scenes, we documented rich avifauna comprising over 40 genera, distributed across trophic levels and ecological niches. Notably, wetland habitats had highest species diversity, followed by early-successional agricultural fields transitioning to grassland systems. We detected several species previously undocumented in the exclusion zone, particularly waterfowl and migratory species that may not have been present during past surveys. In addition to documenting bird species presence, we used acoustic indices within each habitat class to explore aspects of biodiversity ecosystem health not captured by bird species counts. This study provides a snapshot of avian community composition during this unique post-abandonment and pre-remediation window, establishing a reference against which future ecological assessments can be measured. We intend these findings to contribute to our understanding of ecosystem health following catastrophic disturbance and to provide context for future remediation planning.

P2-15

Signature whistles for occurrence, recapture and density (SWORD) for Indian ocean humpback dolphins, *Sousa plumbea* in South Africa

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Baseline data on population density and spatial use are vital for effectively conserving wild populations, but can be challenging for elusive species. Passive acoustic monitoring (PAM) is a cost-effective and non-invasive method that has proved valuable, particularly for elusive species, and is well-suited to dolphins. As dolphin species develop individual signature whistles to convey identity, PAM can be conducted at the individual level. A novel framework called SWORD (signature whistles for occurrence, recapture, and density) targets these whistles for individual detection, which could be critical for species like Sousa plumbea with low encounter rates. To evaluate SWORD's efficacy, a passive acoustic array was deployed in high-use bays within the South African range of S. plumbea. The deployment took place over 42 days, utilising five hydrophones to monitor a portion of the species' southern distribution. The acoustic array detected individuals using automated species identification and bout analysis (SIGID). During the deployment, 1282 total signature whistles from 25 different signature whistle types (SWTS) or individuals were detected. Capture histories based on whistle detections were analysed using capture-recapture models (POPAN JCS) and spatial density models (SECR). SECR models effectively estimated population density, aligning well with previous capture-recapture estimates from localised photo ID studies at various scales. The effectiveness of the SWORD framework underscores its potential as a monitoring tool for elusive, endangered coastal dolphin species through individual acoustic detection. These findings emphasise the importance of PAM in understanding endangered species populations, providing valuable insights for conservation management practices.

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P2-16

Evaluating automated contour extraction methods for bottlenose dolphin whistles

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Reliable extraction of whistle contours is fundamental to advancing bioacoustic analyses of bottlenose dolphins (Tursiops truncatus). Yet, existing automated methods remain constrained by environmental noise, overlapping vocalisations, and dataset-specific tuning that limits broader applicability. This study systematically evaluates existing contour extraction algorithms, spanning both established and deep learning-based approaches, to determine their efficacy under controlled and dynamic acoustic conditions. Leveraging a benchmark dataset of annotated whistles from the Sarasota Dolphin Whistle Database, we assess performance across varying signal-to-noise ratios and investigate the role of pre-processing techniques in improving detection accuracy. Quantitative metrics will provide a rigorous basis for identifying robust methodologies suited for passive acoustic monitoring. By establishing standardised benchmarks and refining extraction workflows, this research aims to enhance automated whistle detection, support comparative studies across datasets, and inform long-term cetacean monitoring strategies.

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P2-17

Passive acoustic monitoring of pollinators: integrating IoT sensors and deep learning for species identification

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Pollinators, such as bees, hoverflies, and butterflies, are excellent bio-indicators of biodiversity, reflecting environmental health. Many species, particularly wild bees and hoverflies, are often classified as Data Deficient (IUCN), highlighting the need for increasing monitoring efforts to gather information and protect them. New technologies can help fill this gap, offering transparency, traceability, and measurable assessments. We aim to develop passive monitoring methods (PAM) focused on Apoidea and Syrphidae by leveraging IoT bioacoustic sensors. These sensors will record insect flight sounds, using machine learning algorithms for automated species identification while minimising the need for specimen capture and sacrifice. Although acoustic identification has proven effective for certain wild bee species, its application across a broader range of taxa remains limited, particularly for hoverflies, for which little bioacoustic data exists. To overcome this limitation, we constructed an acoustic library comprising 549 recordings from 53 species to serve as training data for the identification algorithm. Concurrently, we deployed IoT sensors at six sites in Turin (TO) to record insect buzzing sounds. While these sensors do not independently identify species, the collected audio data are transmitted via 4G connectivity to the cloud for further processing. The classification algorithm, developed using a ResNet50 Convolutional Neural Network (CNN) in MATLAB, has demonstrated promising accuracy in distinguishing species. Additionally, incorporating behavioural variables into the model further enhances classification performance. Once fully integrated into field-deployed sensors, this system will enable large-scale PAM, facilitating the collection of spatially and temporally continuous biodiversity data at an unprecedented scale.

P2-18

Bayesian sound source localisation

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Passive sound source localisation (SSL) is a vital asset in the bioacoustics toolkit, supporting applications ranging from population estimation to long-term territorial analysis. Near-field SSL infers the location of vocalising animals (e.g., birds or bats) by utilising time differences of arrival (TDOAs) estimates between pairs of spatially distributed passive acoustic monitors (PAMs). Like all real-world measurements, TDOAs are subject to errors and uncertainty. Standard approaches estimate the most likely SSL by residual minimisation. That is, they produce point estimates lacking a formal quantification of uncertainty. In their recent systematic review, Rhinehart et al. (2020) identify uncertainty quantification, robustness, and open availability of tools as core challenges to greater adoption of SSL in ecological research. In this work, we present a Bayesian approach to near-field sound source localisation, effectively tackling the uncertainty quantification and robustness challenges. Leveraging the expressive power of Markov chain Monte Carlo (MCMC), we implement a statistical model that incorporates uncertainty as a result of TDOA estimation, microphone placement, environmental effects, and outliers. The resulting tool effectively captures measurement uncertainty, producing a principled probabilistic estimate of sound source localisation. By making localisation uncertainty explicit, this work promises to empower ecologists with the tools to draw stronger, more defensible, conclusions from bioacoustic data.

P2-19

Wide diversity of anthropogenic noise dominates in the biggest French freshwater lake soundscape: monitoring and possible implications

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Anthropogenic noise is a widespread but less visible symptom of human activities around the world. Its ecological impact and assessments on fresh waters remain under-investigated. French peri-alpine lakes are key resources in these territories. However, the development of tourism imposes an increase in noise pollution and related environmental issues. This study aims to characterize long-term spatial and temporal variations of soundscape in the largest freshwater lake in France, Bourget Lake (> 40 km2), using passive acoustic recordings. Passive acoustic recordings took place at five strategic locations that differ in terms of noise exposure between May 2021 and October 2022 in Lake Bourget, the largest natural lake in France. We compute acoustic indices, sound levels and used unsupervised automatic approaches to characterize the diversity and variation of artificial sounds in space and time. Most variations in sound levels can be related to anthropogenic activities, while Biophony and Geophony were minor contributors to this lake's soundscape. At low frequencies (mostly below 70 Hz), the soundscape is dominated by an anthropogenic sound of unknown source that produces sounds mostly on weekdays with a well-defined daily pattern. Boating was the major contributor to the soundscape, with a peak activity during the summer and a daily pattern consistent with tourist use of the boats. This study also reports how anthrophony can include a wide range of temporal and spectral features. The high variation and unpredictability of anthropogenic noise might have important implications for the effects of noise on aquatic life.

P2-20

Lightweight associative memory AI models for bioacoustic event detection

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Here we present first-of-their-kind associative memory AI models for bioacoustic event detection based upon Hopfield neural network architecture. The models are transparent, lightweight and fast-to-train with accuracy scores that compare well with commercial classifiers and convolutional neural networks (CNNs). Currently CNNs dominate academic research into automated solutions to the data processing backlog generated by the adoption of passive acoustic monitoring (PAM). Our models help to alleviate this backlog and accelerate the conversion of data to wildlife conservation insights and to the assessment of welfare and environmental interventions in captive settings. No computationally costly transformations to spectrograms or large labelled datasets for training are required, with a standard laptop, running a model in the background, able to make 340 detections/classifications per second and process over 5.5 hours of raw sound files per minute. The models have broad applicability, from uncluttered ultrasonic soundscapes and the identification of bat echolocation pulses to busier lower frequency ranges and primate vocalisation detections. We demonstrate that they can be built swiftly for new species and locations, whilst retaining accuracy scores of over 80%. A working model can be running within 24 hours of receiving a new dataset, a rapid data-to-insight turnaround especially essential when monitoring welfare in captive settings. Additionally, these fast lightweight models show great potential for the analysis of real-time data on edge-processing devices.

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P2-21

Using acoustic data to study the spatiotemporal distribution of Minke whales in the southwestern range of the North Atlantic subspecies

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Balaenoptera acutorostrata – the common minke whale or northern minke whale – is the smallest species in the Balaenopteridae. This species had a broad geographical distribution in the North Atlantic Ocean, but its seasonal movement patterns remain largely unknown. Minke whales are difficult to locate and identify from a distance due to their inconspicuous and small blow. Most Passive Acoustic Monitoring (PAM) work on the North Atlantic subspecies, Balaenoptera acutorostrata acutorostrata, has focused on their detections in the spring and summer in higher latitudes, where they have been observed as far north as Baffin Bay, Franz Josef Land, and Novaya Zemlya. In contrast, their winter distribution in the Caribbean Sea, the southern extent of their range, remains comparatively understudied. Through a large collaborative effort, this study aims to assess the presence of this subspecies in the southwestern range of its distribution. To achieve this, a Convolutional Neural Network (CNN) detector was used to identify the low-frequency (< 1 kHz) pulse trains produced by the North Atlantic subspecies in previously collected long-term PAM datasets. The results provide temporal information (both diel and seasonal) on the presence of vocalizing individuals, which we aim to correlate with environmental factors (e.g., temperature, depth, and chlorophyll concentration). Understanding these patterns is essential for gaining insights into the species' behavior and habitat use. It will help better assess regional variability and its implications for cetacean bioacoustics, ultimately supporting more effective conservation strategies in the face of increasing human impacts on marine environment.

P2-22

IPI analysis for passive acoustic identification of sperm whale units

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Sperm whales (Physeter macrocephalus) live in stable social units of adult females and their offspring, forming the basis of a complex, multi-level social structure that includes long-term associations and cultural groupings known as vocal clans. Monitoring these units is vital for understanding behavior and ecology, but visual identification is often limited by environmental and logistical constraints. To address this, we explored a passive acoustic method for unit-level identification. Sperm whales produce broadband echolocation clicks with a multi-pulse structure. The inter-pulse interval (IPI) between the first and following pulses correlates with the size of the spermaceti organ and thus body size. Previous studies show IPI can distinguish individuals of different sizes. Because each unit comprises individuals with unique size compositions, IPI distributions may serve as acoustic signatures for identifying social units. We analyzed a long-term acoustic dataset from Dominica, using two types of recordings: targeted first-clicks from known individuals during foraging descents, and systematic recordings from encounters with only one known unit. Kernel Density Estimation (KDE) was applied to the first-click data to create unit-specific IPI distributions. Each daily dataset from the systematic recordings was then compared to these KDEs to estimate its likelihood of originating from each unit. Our results show IPI-based classification reached up to 78.26% accuracy using a lognormal kernel with a 0.01 bandwidth. These findings suggest that IPI distributions offer a promising non-invasive acoustic method for identifying well-documented social units, enabling continued monitoring when visual identification is not possible.

P2-23

Autonomy and environmental impact of passive acoustic monitoring systems: Insights from interviews with practitioners

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Technological progress in acoustic recording devices has made significant strides, providing greater convenience and usability. However, several key limitations still need to be addressed. While these sensors benefit from a great deal of autonomy, they still rely on human intervention to a considerable degree. Indeed, questions on how long these sensors can reliably operate independently to acquire and store audio, often guide and constrain sampling efforts. As research in bioacoustics and ecoacoustics grows dependent on these devices, and as their use and production rise, investigating their environmental impact becomes more pressing. Considering that battery waste and travel emissions are major concerns in this regard, it becomes clear that the notions of environmental impact and autonomy in the case of acoustic sensors are interlinked. This interconnection suggests value in a joint study of both aspects. To explore this further, we conducted interviews with practitioners in bio- and eco-acoustics that cover current practices, limitations related to autonomy and environmental concerns. Drawing from insights from these interviews, we discuss the outcomes of choices in survey design, deployment, and maintenance, as well as existing mitigation strategies. Through this work, we highlight how factors such as deployment location and duration, recording schedule, and the types of sensors and batteries affect both autonomy and environmental sustainability, and how these aspects are interconnected. We also initiate a discussion on what needs to be addressed, through either technological innovation or alternative practices, to achieve a more sustainable and autonomous approach to acoustic monitoring.

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Echoes in the dark: How land use and land cover (LULC) shapes nocturnal bird community in a tropical landscape, Nigeria

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Understanding the effects of land use and land cover (LULC) on bird communities is essential for biodiversity conservation in tropical landscapes yet most studies in this region focus on diurnal species. This study assessed how different LULC types—savannah, gallery forest, rocky outcrops, farmland, and residential areas—influence the assemblage, richness, and distribution of owls and nightjars in and around the Amurum Forest Reserve (AFR), Nigeria, using Passive Acoustic Monitoring from November to December 2021. Species richness was significantly higher in protected habitats, particularly savannah and gallery forests, compared to unprotected residential areas, which exhibited the lowest richness and activity. Farmlands also supported considerable richness, suggesting their potential as supplementary habitats. Residential zones were negatively associated with nocturnal birds, likely due to habitat degradation, anthropogenic disturbance, and cultural persecution. These findings underscore the importance of protected areas, habitat heterogeneity, and sustainable land-use practices in conserving nocturnal birds. Conservation efforts should integrate habitat protection, public education to address negative perceptions, and land management strategies that promote species coexistence. This study provides valuable insights for conservation planning, emphasizing the need for habitat connectivity, heterogeneity and biodiversity-sensitive land-use policies in tropical Africa.

P2-25

Long-term bioacoustics monitoring of birds with deep learning: Patterns of vocal activity in the Hula valley

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Recent technological advancements have enabled the large-scale collection of bioacoustic data, with autonomous recording units continuously capturing vast numbers of bird vocalizations in natural habitats. In this study, we aimed at developing a framework and database that leverage audio signal processing and deep learning models to automatically identify bird species based on their vocalizations. A structured database has been implemented to store short bioacoustic segments along with corresponding annotations and metadata, facilitating detailed analysis of species-specific vocal patterns and broader ecological trends. The system is designed to support long-term monitoring of avian populations at high spatial and temporal resolution, enabling the detection of changes in vocal activity associated with climate variability, disease outbreaks, and anthropogenic noise. Over the past five years, recordings were collected from multiple autonomous recorders deployed across the Hula Valley, a major migratory stopover. A training dataset was constructed comprising approximately 100 resident and migratory bird species, each represented by several hundred manually annotated and filtered audio segments. Deep learning models were trained or fine-tuned using the pre-trained BirdNET model with linear probing. While high precision and recall were achieved for certain species, performance remains suboptimal for others under field conditions. To visualize vocal activity, we developed the 'species vocalogram', 'intensity vocalogram', and 'species richness vocalogram', which capture seasonal and diurnal patterns across multiple species. Long-term analysis of recordings from Agmon Hula Lake Park revealed notable findings, including changes in vocal activity associated with an avian flu outbreak and interannual patterns in species phenology and richness.

P2-26

Articulatory control in a harbor seal: Front-back movement of the tongue affects formant production

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Formants, the resonance frequencies of the vocal tract that filter the sound produced in the glottis, are a keystone of mammalian vocal communication, and can convey information about the caller's sex, size, and identity. However, we still don't know much about how non-human animals actively manipulate formants, a process called articulation in human speech. We trained a harbour seal to produce an airborne vocalization with an oscillating formant structure. This vocalization consisted of several repetitions of a sound resembling a speech syllable "wa". From audio recordings, we estimated the changing frequencies of the first two formants. Simultaneous video recordings allowed motion tracking of the seal's jaws, and an ultrasound scan taken from under the seal's chin in the sagittal plane allowed us to track changing tongue shape. These synchronized data streams were compiled into time series containing formant values, jaw opening, and tongue shape at each time point. A principal component analysis revealed that 87% of the observed tongue shape variation could be captured by only two principal components, corresponding to the up-down and front-back movements of the tongue. The up-down movement matched the opening/closing of the jaws and changes in the first formant. The front-back movement correlated with the second formant frequency of the vocalization, analogous to the front/back dimension in a vowel chart of human speech. Our results show that a harbor seal can dynamically modulate formants when opening and closing its mouth during vocalization, and also displays considerable articulatory control over its tongue.

P2-27

Tokay gecko (*Gekko gecko*) male mating calls are expiratory vocalizations extended by vocal-fry-like pulses

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The tokay gecko (Gekko gecko) produces loud and structured advertisement calls, that serve territorial and mating functions. The advertisement call sequence consist of several rattle and the characteristic To-Kay bi-note call. While the acoustic properties of these calls are well documented, the physiological mechanisms driving their production remains poorly understood. In this study, we investigate the coordination between respiration and phonation in tokay geckos by analysing publicly available video footage with deep learning-based pose estimation (SLEAP). We show that all calls in the sequence are made exclusively during expiration. The To and Kay notes are interspersed by ~1ms short mini-breaths, as is also observed in many songbird species. Furthermore, our data show that these mini-breaths allow extension of call duration by adding discrete, single sound pulses, which suggest gecko vocal folds vibrate in the vocal fry register. Because Tokay geckos are not able to change their source level, e.g. via the Lombard effect, our data suggest that call duration and therefore call energy can be increased by exploiting this vocal register at minimal energetic cost.

P2-28

Sensory-motor control of biosonar output levels as a function of target range, target strength, and masking noise in harbour porpoises (*Phocoena phocoena*)

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For echolocating toothed whales, the combination of varying ranges and target sizes means that received echo levels for a constant source level (SL) will fluctuate over a range of more than 100 decibel, which greatly exceeds the functional dynamic range of the auditory system and thereby impedes acute echo processing required for successful biosonar-guided prey interception. Toothed whales have been proposed to solve that problem by employing range-dependent automatic gain control (AGC) on both the transmitting and receiving side apparently to stabilize perceived echo levels. Here we trained two harbour porpoises to actively intercept targets of different sizes in different noise conditions to test the hypothesis that they employ closed-loop vocal motor control, where click outputs are adjusted to the level and delay of each preceding echo. We reject this hypothesis by showing that SLs are reduced in a range-dependent manner that is mostly independent of target size. Therefore, porpoises do not seek to stabilize perceived echo levels in the auditory system but adjust their SL with range, so the echoes arrive at low levels across a ~30 dB dynamic range. We also demonstrate that exposure to masking noise leads to both increased clicking rates and SLs for the same target ranges via a Lombard response that only partially compensates for decreases in echo-to-noise ratios. We conclude that echolocating porpoises actively seek to generate weak echo levels via range-dependent SL adjustments, and speculate that they likely gauge target size from a combination of echo level, delay and SL.

P2-29

Stable sensory sampling in highly unstable sensory scenes: Foraging toothed whales use highly stereotyped echolocation behavior in the wild

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The ability to extract sensory information from the environment and using it to efficiently acquire food is fundamentally important for survival. Toothed whales have evolved the ability to target and intercept prey in darkness at depth using echolocation. To do so successfully, they must keep careful track of prey echoes over 10s of meters by measuring the time-delay between their outgoing clicks and the returning echoes. Initial experiments on captive animals suggest that part of this timekeeping occurs on the transmitting side, with interclick-intervals (ICI) being dictated by the two-way travel time of sound and a 20ms lag-time, predicting a uniform ICI distribution. This is at odds with the classic modes of echolocation, which split prey capture attempts into the search-, approach-, and buzz-phases, predicting highly modal ICI distributions. Here we tested the hypothesis that toothed whale biosonar sampling is driven by the two-way travel time plus 20ms lag-time by analyzing tag data from 5 toothed whale species that span 4 orders of magnitude in body mass. We show that the biosonar of toothed whales is highly modal, with stereotyped ICI rates for the search-, and buzz-phases of echolocation. The operating modes are species-specific and correlate with body mass, suggesting it is driven by biomechanical resonance and longer detection ranges from more powerful biosonars. We therefore reject the hypothesis that the ICI distribution employed by toothed whales is uniform and conclude that they have preferred ICI operating modes when targeting and intercepting prey.

P2-30

Biosonar ambiguity function : a comparative approach of the complex bat versus the powerful sperm whale click train strategy

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Echolocating mammals such as bats and toothed whales have been using waveform diversity for over 50 million years. This allows them to live in conditions that do not necessarily require high visibility. The hypothesis is that biosonar has evolved independently in bats and sperm whales, optimizing echolocation performance in air and water environments. Nevertheless, many biosonar parameters during navigation or foraging are similar between animals in the two environments. However, it is difficult to know exactly what the animal's intentions are when using echolocation. Here we show that the signal processing techniques used for radar and sonar systems can be applied to estimate the performance of echolocation signals. The wideband ambiguity function provides a time delay and Doppler compression factor representation, revealing less complex acoustic modulations in sperm whales than in bats. In the context of foraging or navigation, sperm whales tend to emit only simple, intensity, and frequency modulated clicks. In contrast, bats can dynamically modify their call parameters in both time, frequency, and intensity to achieve different goals. This method is not widely used to understand the hunting and communication strategies of echolocating mammals. Therefore, the method is currently being applied to other toothed whales, to gain a broader understanding of biosonar patterns and strategies.

P2-31

A sad song: The consequences of developmental stress in songbirds

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Ongoing exposure to chronic stress in the postnatal developmental period can cause detrimental physiological and behavioural changes that can persist into adolescence and impact fitness and well-being throughout life. In humans, postnatal developmental stress, both in the form of sudden trauma and chronic exposure, has been linked to speech impairments and speech delay. Similar stress-related effects on vocal communication have also been observed in the zebra finch (Taeniopygia guttata), the most widely used animal model to understand human speech learning and production. Subsequently, peak vocal performance is a widely recognised indicator of fitness in zebra finches. Peak vocal muscle performance is necessary for peak song performance, which is only obtained with daily vocal exercise and practice. This study explores whether the known changes in vocal perfor-mance due to postnatal developmental stress can be explained by changes in muscle performance. The link is investigated by testing whether interference from experimentally introduced noise pol-lution can affect peak song performance and influence underlying physiological properties due to a reduction in vocal exercise and training. The aim is to uncover the mechanism for how postnatal developmental stress affects vocal learning and song production, and by that expand our under-standing of how developmental stress shapes song production.

P2-32

Acoustic behaviour in bottlenose dolphins during two different target discrimination tasks

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Echolocation is an active sensory system that gives an acoustic representation of the surroundings by the animal emitting clicks, detecting and analysing echoes. However, little is known about dolphin biosonar in moving animals, as well as of the process of controlling click emissions, and the cues used by the animal to discriminate between different targets. We studied how bottlenose dolphins may use their dynamic sound production and hearing abilities, along with head and body movements, to detect and classify objects through echolocation. We designed two different experiments with blindfolded bottlenose dolphins (Tursiups truncatus) swimming freely along the pool and actively discriminating between two targets of three different materials, and a second one to determine the capability to discriminate wall thickness differences in cylindrical targets of the same material. By means of synchronized cameras and hydrophones we observed clear differences between individuals of the same species in terms of ability to discriminate, but similar performance with the harbor porpoises in relation to the materials chosen with the difficulty of the task. Studying how the dolphins solved a simplified echolocation task, allow a better understanding of the processes behind target detection and discrimination capabilities during natural biosonar circumstances.

P2-33

Fundamental and Dominant Frequencies in Bird Vocalizations: A Comparative Analysis

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The production of vocal sound in birds is highly diverse and shaped by the morphological variability of the syrinx, their unique vocal organ, and the vocal tract. This study investigates the relationship between fundamental frequency (f0) and dominant frequency (DF) in bird vocalizations, aiming to clarify the correspondence between these two key acoustic parameters. Although f0 and DF often are the same, they differ in a significant number of species, indicating that they cannot be used interchangeably and should not be conflated. Two datasets were analysed: (1) vocalizations (songs and calls) from 100 Passeriformes species and (2) calls from 100 species spanning 20 avian families. Using Fourier-based and autocorrelation-based methods, both f0 and DF were extracted and compared across species; in addition, vocalization types (songs vs. calls) were compared within Passeriformes. In Passeriformes, songs and calls exhibited similar mean f0 values, while DF differed significantly, confirming that f0 and DF reflect distinct aspects of vocal production. In the broader avian dataset, DF was significantly higher than f0 across most families, although overlap between the two measures varied by taxon. Morphological traits such as body mass and beak size were included to explore potential physical influences on frequency parameters. Additionally, the influence of phylogenetic relatedness on acoustic parameters was considered. The findings highlight that DF and f0, while often related, are not interchangeable measures. A clear distinction between these parameters is essential for accurately interpreting the acoustic signals of birds, particularly in studies linking vocal traits to biological or ecological functions.

P2-34

Quantifying Myonuclei in Zebra Finch Syrinx Muscle Fibers

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Songbirds are widely accepted as the closest animal analogue for human speech acquisition and are therefore extensively studied. There are numerous parallels between human speech learning and song learning in the songbird zebra finch (Taeniopygia guttata), making the zebra finch an ideal model organism to study imitative vocal learning. Song production in songbirds is controlled by the syrinx, which relies on specialized syringeal muscles. Syringeal muscle cells, like all other skeletal muscle cells, are unique by being multinucleated. However, little is known about how myonuclear quantity influences vocal performance and overall fitness. The myonuclear domain hypothesis suggests that each myonucleus governs a defined cytoplasmic volume. As such, hypertrophy would require the addition of myonuclei, and atrophy would result in their loss. While this model provides a framework for understanding muscle plasticity, recent studies have challenged its universality, making it a topic of ongoing debate. Zebra finches must sing daily to maintain peak vocal performance, as singing prevention leads to, amongst other things, muscle atrophy, highlighting the importance of maintaining muscle size. This study aims to determine whether myonuclear quantity in syringeal muscle fibers serves as a predictor of fitness in zebra finches. To achieve this, individual muscle fibers will be enzymatically dissociated using collagenase. Quantification of myonuclei will then be performed using quantitative polymerase chain reaction (qPCR). By examining the relationship between myonuclear number and vocal performance, this research will contribute to a deeper understanding of the neuromuscular adaptations underlying song production and their potential implications for avian fitness.

P2-35

Rapid sensing of approaching objects by horseshoe bats through Doppler shift detection

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How do bats perceive moving objects? To address this question, we conducted a phantom echo experiment with Japanese horseshoe bats (Rhinolophus ferrumequinum nippon). Echolocating bats use echo delay to estimate target distance and Doppler shift information to detect relative velocity. However, how they perceive moving objects remains unclear. To investigate this question, we developed a real-time virtual echo presentation system and conducted phantom echo experiments with bats perched in a flight room. Echoes from an approaching object are theoretically characterized by both changes in echo delay and the presence of Doppler shift. We, therefore, reproduced an artificial approaching object by encoding these two acoustic features into the virtual echoes. Our results showed that only Doppler shift evoked escape response, indicating that they rely solely on Doppler shift. This strategy is particularly advantageous for forest-dwelling horseshoe bats, which inhabit cluttered environments rich in background echoes. In such conditions, reliance on echo delay could introduce ambiguity, whereas Doppler shift, which directly encodes target velocity, provides a more robust and reliable cue. Moreover, because bats can respond within a single pulse, this strategy enables rapid reactions to approaching objects. In fact, their immediate response to Doppler shifts demonstrates that they can detect and react to approaching objects within a single pulse. Consequently, this strategy is likely to provide an efficient mechanism for the rapid and reliable detection of danger in cluttered forest environments.

P2-36

Using machine learning to segment zebra finch song

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Male zebra finches (Taeniopygia guttata castanotis) need daily singing exercise to main their peak vocal performance, which is decisive in competing for territories and finding a mate. However, it is not known how quick the males recover after not singing for a while and how long it takes for them to reach their peak vocal performance again. Investigating this requires monitoring and analysing the singing activity of many birds for weeks and big datasets like this are unfeasible to process manually. To solve this, I trained a machine learning model to automate song segmentation. I created a generalised song model, trained on 7 birds, to recognize zebra finch song. The dataset consisted of 8 birds, with one bird withheld from training to evaluate the song model on unseen data. Additionally, I created specific syllable models for each bird. All models are deep learning models based on the TCN method. I used the DeepAudioSegmenter (DAS) software to annotate all audio recordings, train the models and make the predictions. With these models, I will investigate the singing recovery of 8 singing prevented male zebra finches.

P2-37

A physical model of the zebra finch syrinx

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Songbirds produce highly variable, elaborate songs critical to their communication and survival. However, we still have a very minimal understanding how sounds are made and controlled by their vocal organ, the syrinx. Physical models can greatly enhance the mechanistic insight in the function and control of complex systems. To understand sound production mechanisms in songbirds, we propose a physical model of zebra finch syrinx. Previous studies utilized simplified models, which represented the medial vibratory masses (MVM, i.e., the medial labium and medial tympaniform membrane) as a latex membrane mounted on a tube. Here we developed a physiologically more realistic model based on X-ray CT scan data of the zebra finch syrinx. High-speed measurement of the flow-induced vibrations of the syrinx model revealed oscillation characteristics of the MVM. Our results indicated that adduction of the lateral labium is the key to inducing vibrations of the medial labium.

P2-38

Vocal activity of European songbirds on their wintering grounds in Afrotropical mountains

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Acoustic communication plays a vital role in the lives of birds, allowing them to maintain contact within a group, warn each other of danger, defend their territory, and attract mates. While the functions of song in European bird species are well understood, their vocal activity outside the breeding season remains poorly studied. This is particularly surprising given that an estimated 62% of birds migrating from the Palearctic to Africa also sing on their wintering grounds. In this study, we analyzed the vocal activity patterns of the Willow Warbler (Phylloscopus trochilus), Wood Warbler (Phylloscopus sibilatrix), Pied Flycatcher (Ficedula hypoleuca), and Garden Warbler (Sylvia borin) wintering in the Bamenda Highlands of northwestern Cameroon. To achieve this, we used recordings collected through automated acoustic monitoring over the course of one year. Our findings indicate that the Willow Warbler was the most vocally active species, especially toward the end of the wintering period, with peak activity in February. During this time, Willow Warblers produced more songs than calls and sang more in the morning than in the late afternoon. This pattern suggests a functional connection to the approaching migration and the upcoming breeding season. The other studied species exhibited significantly lower vocal activity. By deploying six autonomous recording units, we strengthened the robustness of our findings and gained a more detailed understanding of temporal vocal patterns. These results contribute to filling a knowledge gap regarding bird behaviour outside the breeding season.

P2-39

Individual identification and call type classification with unsupervised learning

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Individual identification and call type classification are crucial for many ecological tasks, such as tracking individuals, classifying behaviour and estimating population. Recently, deep learning has proved to be a promising tool for bioacoustics, particularly for species classification, alleviating the need for manual annotation of long passive acoustic monitoring audio recordings. However, in the context of individual identification, there is little available data labelled at the individual or call level, restricting the applicability of supervised data-driven approaches to a few species and datasets. One approach is to segment the calls as sequences of syllables to facilitate individual ID or call type classification (TweetyNet, Cohen et al., 2022), but still relies on per-species syllable annotations. In this work, we build a simple and efficient, fully unsupervised algorithm to decompose any call from any species into a sequence of syllables. We first separate the notes based on silences (Sainburg et al., 2019), then use hierarchical clustering to group these notes into clusters and extract templates -syllable representations- before performing matching pursuit to decompose each call as a sequence of syllables. We then explore whether this approach can help identify an individual's unique vocabulary and vocal signature. We investigate whether the intra-species, per-syllable variability is sufficient for clustering and classifying individuals.

P2-40

Big whales and big data: a convolutional neural network approach to investigating long-term spatial-temporal patterns of blue and fin whales in the Southern California Bight

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Blue (Balaenoptera musculus) and fin whales (Balaenoptera physalus) produce low-frequency calls associated with foraging, social interactions, and reproduction. Sonobuoys deployed during California Cooperative Oceanic Fisheries Investigation (CalCOFI) surveys contain recordings of distinct calls produced by these species; namely blue whale A, B and D calls, and fin whale 20 Hz and 40 Hz calls. Here, we describe a convolutional neural network approach to automatically detect and classify these calls in spectrograms, enabling subsequent analysis of long-term patterns. We fine-tuned the object detection model Faster-R-CNN with a dataset of 10,072 labeled calls collected during seven CalCOFI cruises spanning numerous sonobuoy deployments. We reserved data from four CalCOFI cruises spanning each season for testing. Mean average precision (mAP) across all categories was 0.82 on the testing data. The model performed well on blue whale A, B, and D, and fin whale 20 Hz calls (average precision 0.80–0.93), while fin whale 40 Hz calls were more challenging (0.71). We deployed the best performing model on long-term (2004-2025) CalCOFI recordings to investigate spatial-temporal patterns in blue and fin whale acoustic activity in the Southern California Bight. Seasonal peaks were consistent with previous studies, and long-term trends showed an increase in the number of blue whale calls detected per survey over the study period, while fin whale call detection rates remained stable. During the 2014–2016 marine heatwave, calling activity declined for both species, with blue whale calls largely restricted to the shelf break, suggesting habitat compression. This work illustrates the value of integrating passive acoustic monitoring and deep learning to investigate ecological patterns and track species responses to environmental change.

P2-42

Learned and un-learned responses to anti-predator vocalizations in a juvenile bird

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Anti-predator vocalizations and responses are key for animals to survive, especially when they are naïve and vulnerable young. Being able to respond appropriately to a novel threat is important, however, being able to flexibly adapt to new predator conditions can also be important. To better understand how these trade-offs may be managed by call type, we examined the response of naïve great tit fledglings to different types of anti-predator vocalizations and behaviours to determine if call type/context impacted their response. We found that responses to calls signalling an immediate and lethal threat appeared unlearned, while calls signalling a less immediate thereat did not induce the anti-predator behaviour in fledglings as it did in adults. This suggests that different types of predator contexts and anti-predator calls have different types of acquisition, allowing for both immediate unlearned response in high threat contexts, but flexible and delayed responses in low threat contexts for naïve individuals. This work was funded by University of St Andrews 600th Anniversary and St Leonard's Fee Scholarships.

P2-43

Novelty and constraints during the evolution of vocalization in tetrapods.

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All vocal tetrapod lineages (anurans, lepidosaurs, testudines, mammals and archosaurs) evolved vocal communication between 250-100 mya, as supported by phylogenetic trait analysis of extant species, vocal fold development and the independent evolution of tympanic ears. However, we know very little on the evolution of sound production. Producing voiced sound with vocal folds (VF) requires driving air pressure, a sound source consisting of VFs that exhibit self-sustained airflow driven oscillations, and a vocal tract for filtering and radiation. Both the vocal organ and the vocal tract consist mostly of soft tissues or cartilage that rarely ever fossilize. It thus remains challenging to determine when and how voice production emerged. Drawing on extensive literature describing hyolaryngeal and respiratory structures, we here investigate when key traits linked to vocalization emerged in the major lineages, including modifications to the hyolaryngeal apparatus and respiratory anatomy. Furthermore, we studied scaling constraints on key parameters related to voice production by vocal folds, such as their elasticity and length. We found that within amniotes vocal fold length correlates strongly to body mass. In contrast, vocal fold elasticity does not correlate with body mass but exists in a surpassingly narrow range.

P2-44

Evolutionary mechanisms of acoustic communications driven by reproductive inference in Rana tagoi, Japanese brown frogs

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Reproductive interference is the reduction in fitness that arises when heterospecific populations coexist. One outcome of reproductive interference is reinforcement, a process in which natural selection strengthens premating isolation to reduce maladaptive interspecific mating. In sympatric areas, reinforcement can lead to more discriminating female mate choice, thereby enhancing sexual isolation. Reinforcement is considered a potent evolutionary force that promotes signal divergence and potentially initiates speciation from allopatric populations. However, its mechanisms and evolutionary consequences remain poorly demonstrated. Rana tagoi is a ranid frog endemic to Japan, which has sympatric populations with closely related species, R. kyoto. A previous study indicated that reinforcement led sympatric female R. tagoi to distinguish conspecific calls by species-specific differences in the number of notes in a call. In this study, I compared calls from sympatric areas with those from allopatric areas to demonstrate that reinforcement of female mate choice drove divergence in calls. Additionally, I conducted playback experiments to test whether the divergence in female mate choice or in calls in sympatric areas leads to incompatibility in reproductive communication between allopatric populations. The results showed that divergence in an acoustic trait aligns with the conspecific recognition cues used by females and that the direction of divergence enhances differences between species. The playback experiments revealed that allopatric R. tagoi females preferred the calls of sympatric R. tagoi males, whereas sympatric females did not prefer the calls of allopatric males. This suggests that divergence in female mate choice, rather than signal divergence, induces conspecific incompatibility between populations.

P2-45

A phylogenetic analysis of the evolutionary history of toothed whales' tonal sounds repertoire: a preliminary analysis

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Recent studies show that anatomical specializations may allow toothed whales to exploit lateralized register vibrations to produce a rich repertoire of signals. Combined with their vocal learning abilities, toothed whales' communication is potentially one of the most complex in the animal kingdom, yet the macroevolutionary process driving this complexity remains poorly understood. This is largely due to the lack of a comprehensive and standardized characterization of their vocal repertoire limiting our ability to test hypotheses about factors driving speciation, signal evolution, and vocal adaptations to the environment. In this study, we quantitatively assess the tonal sound repertoires of 15 species representing major lineages of toothed whales. Species acoustic data was obtained from NOAA and the Watkins database. Tonal sound contours were extracted and a total of 200 contours were randomly selected for each species to generate metrics of diversity (Hills numbers), composition (contour type), complexity (modulation), and acoustic structure (frequency and duration). Trait reconstructions were conducted using Maximum Likelihood methods to identify patterns of similarity and divergence of tonal sound traits across species. Preliminary analyses suggest that high frequency, diverse, and complex vocal repertoires are concentrated within the Delphinidae lineage, particularly among species that live in fission-fusion societies, where acoustic communication plays a critical role in group formation and cohesion. Additionally, we addressed the potential impact of incomplete data and unequal sampling effort across species on our results. Together, these findings represent an initial step toward reconstructing the evolutionary history of vocal repertoires in toothed whales.

P2-46

Sounds and scents: testing the role of acoustic and chemical communication in Mantidactylus diversification

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In anurans, bioacoustics is well known to play a role in mate recognition, whereas the relevance of chemical communication in the breeding context is still poorly understood. With around 500 mitochondrial lineages, Mantellidae represent the biggest vertebrate radiation from Madagascar. Inside Mantellidae, the males of mantellins, i.e. the richest subfamily in terms of species, are known to produce volatile compounds, which are one of the few studied pheromones in anurans. We hypothesise that both acoustic and chemical communication contribute to their exceptional diversification. We test our hypothesis on the Mantidactylus genus. A set of acoustic traits measured for the advertisement calls, i.e. those vocalisations emitted by the males to attract potential mates, has been compiled for this group. We also obtained measurements of the males' femoral glands, which are responsible for the production of the volatile compounds, and are here used as a proxy for chemical communication. Using this information, we compute Mantidactylus species diversification rate and calculate each variable's evolutionary pattern. We then assess the correlation between species diversification and traits' evolution to test chemical and acoustic communication contribution to the genus's diversification.

P2-47

Global diversity and geography of bird acoustic strategies

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Bird songs represent a complex functional trait critical to avian ecology yet remain underexplored in broad-scale biodiversity analyses. We analyzed approximately 116,000 vocalizations from over 3,000 passerine species using the Modulation Power Spectrum (MPS) to create a high-dimensional acoustic space capturing detailed frequency and temporal modulation patterns. This approach revealed distinct acoustic strategies with clear geographic structuring that diff ers signifi cantly from patterns identifi ed through morphological or taxonomic metrics. Our fi ndings demonstrate that acoustic diversity follows unique biogeographical patterns, providing novel insights into evolutionary constraints, ecological fi ltering, and niche partitioning mechanisms. This work advances our understanding of global biodiversity patterns by highlighting how complex vocal behaviors in birds exhibit functional signifi cance and structured distribution across geographic regions.

P2-48

Cranial allometry meets bioacoustics: how neotenic skull shapes associate with vocal behaviour in companion dogs

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The human brain's globularization likely arose from selection for social tolerance via neural crest development, leading to facial shortening and cranial rounding, both aiding speech readiness. Domestication causes morphological and behavioural changes, such as altered craniofacial ratios and vocal behaviour. Dogs show great diversity in function and appearance, making them ideal for studying whether these changes affect vocal behaviour. We used skull morphometrics (N=408) from high-resolution CT scans to examine skull width and craniofacial index (cranium length/nose length) allometries. A Phylogenetic Generalised Linear Mixed Model revealed a quadratic relationship where small companion and molosser breeds were at opposite ends, indicating that selection for extreme sizes results in proportionately rounder skulls. Smaller bulldog-type breeds deviating from this allometry formed a distinct cluster, suggesting a separate selective push due to the baby schema effect towards a neotenic flat face. Using CFA-derived vocal behaviour scores from a questionnaire (N=2544) in zero-inflated linear mixed models, we found that small bulldog-type dogs are less likely to howl at sounds, growl at strangers, and bark in agonistic contexts, but growl more during play, while protecting toys, and when touched by humans. These results suggest that these small bulldog-type breeds obtained their extreme rounded heads and short noses due to selection for neoteny and extreme social tolerance, correlating with lower territorial and defensive aggression and elevated playfulness, both traits favoured in companion breeds. Our findings reveal links between functional selection, morphology and behaviour in dog breeds.

P2-49

Beyond Courtship: Agonistic Sound Production by the Frillfin Goby (*Bathygobius soporator*) from Southeastern Brazil.

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Bathygobius soporator is a benthic, site-attached gobiid species predominantly inhabiting intertidal rock pools and oceanic island habitats across the Atlantic. The frillfin goby was the first gobiid species to have its acoustic repertoire documented—restricted to male courtship displays—by Tavolga (1958) in Florida. Here, we present the first characterisation of agonistic acoustic signalling in B. soporator. Behavioural and acoustic data were obtained under controlled laboratory conditions using three devices: an AudioMoth autonomous recorder; an Aquarian H1a hydrophone interfaced with a Tascam DR-40X recorder; and a GoPro Hero9 camera for synchronised video acquisition. Bioacoustic analyses were carried out in Raven Pro1.6, including measurement of time–frequency parameters and descriptive statistics (minimum, maximum, mean, and standard deviation). Spectro-temporal patterns were visualised through oscillograms and spectrograms, and non-parametric comparisons were performed using Kruskal–Wallis H-tests and corresponding p-values in R v4.4.0. A total of 171 acoustic signals were recorded during dyadic interactions between males, comprising 167 “thumps” and 4 “drums.” Thumps exhibited a mean peak frequency of 1406.2 Hz and an average duration of 0.1 min, while drums had a lower mean peak frequency of 363.3 Hz and pulse counts ranging from 12 to 38, spanning up to 1.8 min in duration. These findings expand upon earlier descriptions by revealing additional signal types beyond courtship-associated thumps and demonstrate the occurrence of vocalisations in agonistic contexts, indicating a broader communicative role for acoustic signalling in this species.

P2-50

First description of 'vocal sparring' behavior between juvenile male sperm whales (*Physeter macrocephalus*) off Mauritius

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The period when juvenile male sperm whales (Physeter macrocephalus) remain within matrilineal units is a key stage in their social development. Investigating their social interactions and vocalisations during this time is essential to understanding socialisation processes and the evolution of communication in the species. In this context, we characterise a previously undocumented socio-acoustic behaviour identified among juvenile males of the 'Irène Gueule Tordue' clan off Mauritius (Sarano, 2022). Data were collected passively and non-invasively through simultaneous acoustic and video recordings. Each click was detected, annotated and assigned to its emitter, even when click trains overlapped, using separation techniques and direction-of-arrival estimation, while also examining specific acoustic features. These methods enabled accurate individual attribution of vocalisations. We introduce the previously undocumented behaviour termed 'vocal sparring': prolonged dyadic interactions, between juvenile males of similar age, involving sustained click trains. The structure of these exchanges aligns with the criteria for play described by Bekoff (1972) and Burghardt (2014): characterised by a lack of aggression, spontaneity and voluntary repetition. Their recurrence across years between the same individuals indicates durable social bonds. This vocalised play behaviour may support the development of social and communicative skills essential to matrilineal sperm whale societies. Our findings provide new insights into juvenile socialisation in male sperm whales and contribute to understanding the evolution of social play and communication in marine mammals.

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P2-52

Effects of playback received level on response strength in free-swimming bottlenose dolphins

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Playback experiments are a valuable tool for studying functions of animal communication signals. However, little attention has been paid to potential effects of received level (RL) on playback responses. To examine this, we analyzed playback experiments of various natural whistle types to wild common bottlenose dolphins (Tursiops truncatus) wearing digital acoustic tags (DTAGs) in Sarasota Bay, Florida. Dolphins were exposed to two broad categories of signature whistle stimuli: 1) familiar (their own whistle, or that of a close relative or associate) and 2) unfamiliar (recorded from a different population). We quantified the overall dynamic body acceleration (ODBA) before and after playback for 43 trials to 20 different individuals. ODBA was found to significantly increase immediately following playback (Wilcoxon signed-rank test, $V = 241$, $p = 0.0001$). This increase was greater following playbacks with higher RL (GLM, estimate = 0.233, t value = 0.314, $p = 0.004$), and females exhibited a greater increase in ODBA compared to males (GLM, estimate = -0.616, t value = -2.935, $p = 0.005$). Duration of elevated ODBA response after playback was also longer following higher RL playbacks (GLM, estimate = 0.039, t value = 2.331, $p = 0.0252$). Louder stimuli likely simulated the sudden, surprising appearance of a nearby animal, and thus triggered stronger responses (as measured by ODBA mean and duration), with stimulus intensity playing a larger role than stimulus type. We suggest using lower source levels to simulate distant animals in experiments aimed at studying signal function.

P2-53

Spatiotemporal analysis of midwife toads (*Alytes obstetricans*) call patterns using a network of spherical microphone arrays.

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During the breeding season, male Midwife Toads (Alytes obstetricans) emit mating calls from land to attract females. From April to August, these calls form choruses of short, tonal sounds. This study aims to describe the spatial and temporal organization of one such chorus near Lyon, France. Initially tested in a semi-anechoic room using playback choruses of green tree frogs, a network of five spherical microphone arrays was deployed in the toads' territory. This work aims, after a geometrical calibration process of the network, to separate and localize the calls within an area of about ten square meters. Detection reports in the temporal domain are then created to compile information on the timing, location, and frequency of specific events. Ultimately, it is possible to have a visual representation of the spatial and temporal distribution of Midwife Toad choruses.

P2-54

Factors affecting song amplitude in six European passerine species

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Amplitude is an inherent feature of every sound and one of the decisive parameters for successful signal transmission. Animals using acoustic signals for communication adjust their amplitude for various purposes related to reproduction, communication with distant recipients, limiting signal range to avoid eavesdropping, and maintaining communication in noisy environments. Despite its importance, signal amplitude has received relatively little attention in studies on acoustic communication. We present calibrated measurements of amplitude, specifically the sound pressure level of songs, across six common European passerine species in relation to temperature, humidity, ambient noise, song length, and song peak frequency. SPL values at a distance of 1 meter were 81 ± 2 dB(A) (75–86 dB(A)) in Common Chaffinch (Fringilla coelebs), 82 ± 3 dB(A) (74–89 dB(A)) in Yellowhammer (Emberiza citronella), 81 ± 1 dB(A) (75–87 dB(A)) in Common Chiffchaff (Phylloscopus collybita), 82 ± 1 dB(A) (77–87 dB(A)) in Willow Warbler (Phylloscopus trochilus), 81 ± 2 dB(A) (74–88 dB(A)) in Common Whitethroat (Curruca communis), and 88 ± 2 dB(A) (83–95 dB(A)) in Great Reed Warbler (Acrocephalus arundinaceus). The generalized linear mixed models analysis revealed that song sound pressure level is significantly influenced by ambient noise levels, temperature, and song length. These findings suggest that environmental factors and song characteristics play a crucial role in determining the amplitude of the species' songs.

P2-55

Neighbours or strangers?: Relationship between geographic and individual variation in yellowhammer (*Emberiza citrinella*) songs

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Geographic variation in birdsongs provides insights into areas like song culture evolution and population dynamics. In species with clear dialects, studies often focus on the dialect part, while more flexible song components are rarely explored. The Yellowhammer (Emberiza citrinella) is a well-known species for its dialects, with its song consisting two parts: a variable initial phrase and a conservative dialect phrase. The dialect is shared among neighbours, and through the citizen science project Dialects of Czech Yellowhammers (DCY), its distribution and types were well documented in Czechia. The initial phrase is variable within individuals, with each male singing a distinct combination, sometimes sharing certain phrase types with other males. Our study aimed to examine the distribution of the shared initial phrase types and their relationship to distribution and dialect regions. To evaluate the sharing frequency between neighbours and distant males, recordings from two localities were used—Vizina, Czechia (N=63), and Swieca, Poland (N=40) — along with the broader dataset from the DCY project (N=180). Phrase similarities were calculated using a dynamic time warping algorithm in Luscinia software, followed by visual inspection from six observers, both experts and non-experts. We found that phase sharing were more common within localities than between far-distanced males, with no clear relationship with dialect regions. Some males shared multiple phrase types with other males, but none had a fully overlapping repertoire. These results support the potential use of songs as individual signatures for acoustic monitoring and provide insights to song learning and cultural dynamics in songbirds.

P2-56

Cultural contact and evolution in nonnative Amazona parrots

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Parrots are highly intelligent vocal learners exhibiting some of the most complex avian communication systems. Due to the pet trade, they are also common as introduced species in many cities around the world; these exotic parrot communities offer an unusual opportunity to study the effects of cultural isolation and secondary contact in avian vocal communities. Two sister species, Amazona viridigenalis and Amazona finschi, have non-overlapping native ranges in eastern and western Mexico, but introduced populations overlap in California, where they socially interact and hybridize. Here, I investigate the effects of introduction and cultural contact on the vocalizations of both species. Comparison with recordings from California and their native ranges indicates that the vocalizations of both species have changed since their introduction and that dialectal variation is emerging within their California range. Furthermore, the two species appear to be converging on shared dialects for certain call types within roosts but differentiating geographically among roosts. I also evaluate the efficacy of random-forest machine-learning techniques in analyzing the complex vocal repertoires of Amazona parrots; these techniques were more successful at clustering wild-type calls than introduced calls.

P2-57

Horned lark intermittent song differences across North America

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Geographic variation in birdsong provides valuable insight into divergence and speciation, as song performance is linked to courtship and mate choice in many bird species. As a widespread species complex, Horned Larks (Eremophila alpestris) provide an opportunity to examine how song variation drives evolutionary differentiation. Horned Larks use their intermittent song type for territory establishment and in flight for courtship. Here we test the hypothesis that 1) intermittent song varies among North American populations of Horned Larks, and 2) this variation is associated with geographic environmental differences. We analyzed song recordings from the Macaulay library and Xenocanto, using Raven Pro and Kaleidoscope Pro to quantify song and soundscape characteristics. Additional field recordings were collected from coastal populations of the Channel Islands and Southern California. We used Raven to measure syllable similarity, entropy, song length, note rate, and frequency range, while Kaleidoscope measured Acoustic Complexity and the Normalized Difference Soundscape Index to capture anthropophony and biophony. We utilized GIS spatial analysis tools to examine the relationship between song variables and climatic factors such as wind speed, habitat type, precipitation, and temperature. Data were analyzed using Linear Mixed Models and Principal Component Analysis in SPSS. Preliminary results indicate wind speed and habitat type are the strongest predictors of song length and entropy in Horned Lark intermittent song. Understanding how song traits are tied to environmental variation has important implications for wildlife conservation as ongoing environmental change is inevitable. Examining variation in intermittent song may provide insights into future differentiation of Horned Larks.

P2-58

Whistle sharing within a family of bottlenose dolphins in Denmark

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Bottlenose dolphins produce individually distinctive vocalizations known as signature whistles, which serve as acoustic identifiers and facilitate individual recognition. However, the function of most other dolphin whistles remains unknown, despite the fact that some are frequently produced in a stereotyped manner. In this study, we examined the stereotyped whistle repertoires of bottlenose dolphins living in Danish waters, outside their species' typical range. This unique situation allowed us to record individuals or small groups using passive acoustic monitoring tools. We recorded five dolphins in total, four of whom were maternally related. These included two solitary male dolphins who were brothers, and a trio of female dolphins: their mother, their sister, and an unrelated female. Our analysis revealed that two stereotyped whistles were shared between the solitary males and the female group. An additional whistle was shared between one of the males and the female group, but not the other male. Each solitary male, as well as the group of three females, also produced one unique stereotyped whistle that was not shared with the others. The presence of shared stereotyped whistles among related individuals suggests the potential use of specific whistle types for group identification, in addition to individual recognition. This study was supported by Aage V. Jensen Naturfond and SDU Climate Cluster.

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P2-59

Contextualised vocal interactions of long-finned pilot whales in the Gascogne gulf (France)

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Long-finned pilot whales (Globicephala melas), as many toothed cetaceans, are both highly vocal and social animals. Their communication systems comprise large vocal repertoires that are learnt and dialectic ; and their matrilinear social groups are hierarchically structured in pods and clans, the former often reaching over hundreds of individuals. These characteristics make the species particularly interesting to study vocal behaviours in social interactions. However, its marine habitat makes data collection challenging. Indeed, to envision a functional analysis of pilot whales' communication system, similar to what has been achieved on several terrestrial species, vocal activities should be associated with particular behavioural contexts. This may be achieved by recording sufficient amounts of simultaneous underwater visual and acoustic data, in which vocalisations can be attributed to individuals. To address this, we propose a relatively low-cost design for a portable video-acoustic antenna. Swimmers can use this system to record both physical and vocal behaviours of aquatic animals, while visually identifying vocalisation emitters. This approach, applied to study spatially stable populations that tolerate human observers, is an opportunity to better understand how marine animals use acoustics to interact. We demonstrate its use with first results on behaviour specific vocalisations and multimodal dyadic interactions, for semi-resident long-finned pilot whales of the Gascogne Gulf (N.E. Atlantic).

P2-60

Aspects of individuality in the song of European nightjars

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The European nightjar (Caprimulgus europaeus) is a cryptic, nocturnal bird renowned for its distinctive song produced during crepuscular and nightime hours, primarily as part of its breeding display. Its song is characterized as a monotonic churring song with a remarkably stable primary frequency and short-duration frequency shifts, known as phrases. Previous studies have revealed that specific acoustic features like pulse rate and phrase length can be used for individual identification. A key requirement for reliable individuality recognition is that the intra-individual variation (variation within the same bird) must be small relative to the inter-individual variation (variation between different birds). This ensures that the natural variability within a single individual's song does not overlap significantly with the differences observed between individuals. In this study, we present preliminary findings on the variability of song parameters within individuals.

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P2-61

Evidence for a graded signal of aggressive intent in male Indo-Pacific bottlenose dolphins

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Aggressive signalling is a widespread mechanism used across the animal kingdom to encode urgency, motivation and fighting ability, and plays an important role in resolving conflicts, preventing escalation to costly, physical contact. During inter-sexual conflicts, males may attempt to coerce females using threat vocalisations to increase their own chance of mating, or to decrease the mating success of others. In Shark Bay, Western Australia, male Indo-Pacific bottlenose dolphins will cooperate to herd single, oestrus females and defend her from rival alliances in events termed 'consortships'. When consorting a female, males produce a threat vocalisation called a 'pop' in rhythmic trains, inducing the female to remain close. In cases where the female does not comply or makes failed attempts of escape, or even when rival males attempt to steal the female, the consorting males may follow pop emissions with the 'crack' vocalisation; an aggressive signal produced with a concurrent rapid horizontal or vertical movement of the male's head (i.e. a head jerk) before potential escalation into physical aggression. Here, we describe the context and coordinated use of pops and cracks in 29 behavioural events and investigate the temporal occurrence of pops before and after the crack vocalisation. Pops are produced at high temporal variability, particularly before compared to after the crack, and while the simultaneous use of pops and cracks appear highly context dependent, we propose that their combined use acts as a signal of aggressive intent.

P2-62

Unlocking hive communication: Exploring acoustic signals in honey bee colonies

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Honey bees (Apis mellifera) exhibit complex social behaviors that rely heavily on effective communication. While previous research has emphasized volatile-based information transfer, the potential role of sound-based communication has received surprisingly little attention, even though honey bees are known to produce a wide variety of sounds. In this study, we quantified acoustic responses of honey bee hives to external stimuli and compared them between two distinct genotypes: "Managed bees," which are widely bred by commercial beekeepers, and "Survivor bees," which naturally occur in Southern California and are known for exhibiting stronger defensive responses to external disturbances. We recorded acoustic responses to stimuli in both genotypes, collecting sound data from entire colonies in the field as well as from individual bees in controlled laboratory settings. Our analyses revealed a consistent increase in both the frequency and amplitude of the sounds produced by hives following disturbance. This increase was significantly greater in survivor bee hives compared to managed bee hives. We also used standardized protocols to quantify hive defensiveness in the field and found that survivor colonies were significantly more defensive than managed colonies. In the laboratory, we observed that individual bees produced distinct acoustic responses when exposed to playback of sounds recorded from disturbed hives. These results suggest that sound-based communication plays an important role in honey bee defense at both the individual and colony levels. Our findings provide beekeepers with a valuable tool to assess hive defensiveness, offering critical information that can improve management decisions and guide selective breeding practices.

P2-63

Songs of the studs: do male song traits influence female choice in a wild, promiscuous bird?

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Acoustic performance often directly influences the reproductive success of vocal animals. Many songbird species have high levels of extra-pair paternity, and extra-pair mating success has been repeatedly associated with male song features. However, few studies have compared female preference for acoustic traits of social mates versus extra-pair mates. Females should prefer social males that signal direct benefits for raising offspring, while preferring extra-pair males that signal indirect, genetic benefits to offspring. We investigated song and reproductive success in wild Savannah sparrows (Passerculus sandwichensis) to understand how female choice is influenced by male song, and how this choice may differ for social and extra-pair mates. We analyzed nestling paternity and songs of social and extra-pair males over five breeding seasons to compare social reproductive success (number of raised offspring) and genetic reproductive success (number of sired offspring) to song structure and performance features. Contrary to our predictions, we found that social and genetic reproductive success are highly correlated. Surprisingly, we found only weak associations between male song features and reproductive success. However, we did detect a relationship wherein both reproductive success and song vary with male age. We discuss the complexity of female choice, and the importance of considering how multiple aspects of individual quality influence reproductive success.

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Sharing the load: Active role of male gerbils in parental communication and behavior

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Socially monogamous species, like the Mongolian gerbil (Meriones unguiculatus), often show biparental care, in which the father aids the mother in raising the young. To investigate how gerbil parents communicate with their offspring and whether this differs between mothers and fathers, we performed reunions with five families at four infant age classes, using an acoustic camera to identify the sender of ultrasonic vocalizations. Additionally, we played pup USVs and ADVs, adult contact calls, and a control stimulus in a playback experiment with ten experienced parents and ten gerbils without pup experience. During the reunion experiments, the removed pups produced USVs and were investigated by both parents. Both parents produced contact calls, but fathers produced more calls than mothers did. Pups did not vary their vocalizations between parents. In the playback experiments, gerbils' responses differed between the stimulus types, and an interaction of sex and experience was found: Male gerbils without prior pup experience responded stronger than inexperienced females. This could be attributed to the similarity of pup USVs to mating calls, or to the novelty of pup calls. However, the reactions of male and female gerbil parents did not differ significantly and no differences in parental communication or behavior were found between the sexes. Together with the findings from the reunion experiments, this confirms the active role that fathers take on in this socially monogamous species.

P2-65

Selective retrieving of own pups by mothers using pups' isolation calls and contact cues in Japanese house bat.

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Maternal care, particularly the retrieval of strayed pups, is vital for the survival of pups. In Japanese house bat, mothers always carry their pups throughout early development, making separation risky. Limited annual reproduction and the physical burden of pup-rearing indicate a high reproductive investment, suggesting a sophisticated system for retrieving. We investigated which sensory cues elicit maternal retrieving. In echolocating bats, which rely heavily on acoustic cues, pups' isolation calls (ICs) are thought to serve as signals that trigger maternal care. We also hypothesize that contact cues play an important role due to close physical proximity between mother and pup. In this study, we conducted two behavioral tests: (1) a presentation test, where pups were directly presented to the mothers, and (2) a playback test, where ICs were broadcast through a speaker. In both experiments, mothers were tested under two conditions: (a) while holding non-presented pups, and (b) with all pups isolated. The results showed that (1a, 1b) mothers retrieved only their own pups during a limited postpartum period; (2a) mothers didn't approach the sound source, whereas (2b) 80 % did, indicating acoustic cues alone can elicit retrieving. These findings suggest mothers are sensitive to contact cues, modulating their motivation to provide care. When contact is absent and motivation is high, mothers recognize their own pups using ICs and selectively retrieve them. Accurate perception of limited sensory cues may help reduce the energetic costs of nurturing.

P2-66

Does nonhuman primate group singing produce rhythmic categories?

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The study of rhythm is increasingly adopting a comparative approach to understanding singing behavior across species and the evolutionary history of human music. Human music is characterized by temporal intervals between unit onsets that produce rhythmic regularities, also known as rhythmic categories. The prevalent rhythmic categories in music are 1:1 (isochrony), 1:2, and 2:1. These rhythmic categories have also been found in the songs of other animal species, particularly in their individual vocal emission during choruses and duets. It remains an open question whether individual rhythmic behaviors, during group singing in many species, may elicit the emergence of group rhythmic structures and how relevant these patterns are for communication. Using the same methodology to investigate individual rhythm, we examined the emergence of rhythmic categories in the collective output of 391 duets of Indri indri, the only singing lemur. We defined song intervals in three levels: a) intervals when at least one individual is singing, b) overlap intervals when both singers emit simultaneously, and c) all unit onsets, considering intervals between adjacent unit onsets. We found isochrony in the collective output of the duets considering all the above levels of analysis. Our results suggest that a natural combination of singers emitting sequences with categorical rhythms 1:1, 1:2, and 2:1 led to the emergence of isochrony in the collective output. Further research might clarify the significance and adaptive value of isochronous output and the selective pressures that drive its evolution across species.

P2-67

Multimodal communication in ants: the interaction between chemical and vibro-acoustic signals in three Myrmicinae species

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Vibro-acoustic communication is found in a wide group of animals, including mammals, birds, and arthropods, the latter being the primary focus of most research. Such signals play crucial roles in several behavioural contexts, from courtship to predator-prey dynamics, being also part of the complex communication systems of social insects. Among social insects, ants predominantly rely on chemical communication. However, some ant species possess a stridulatory organ, enabling the production of vibro-acoustic signals, probably mainly perceived as substrate-borne vibrations. Despite stridulation being described in numerous ant species, the interaction between this mode of signalling and their dominant chemical communication remains poorly understood. Against this background, we examined bimodal alarm signals involving the simultaneous emission of volatile pheromones and stridulations in three Myrmicinae ant species: Crematogaster scutellaris, Myrmica scabrinodis, and Manica rubida. We recorded and characterised both the chemical and the vibrational components, verifying that these ants produce species-specific pheromones and stridulatory signals. Behavioural responses to the chemical, vibrational, and bimodal signals were also tested. Results indicated species-specific variations in multimodal communication, likely reflecting ecological adaptations. For instance, bimodal signals were prominent in M. rubida, while M. scabrinodis displayed a stronger reliance on volatile pheromones. These findings could provide new insights on the role of vibrational signals when combined with other signalling modalities (such as the chemical one), suggesting potential environmentally-driven constraints on the effectiveness of both components in different ecological contexts.

P2-68

Fast trills in songs of a small passerine are not reliable signals of individual's age or predictors of mating success

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Birdsong may convey various information about the singer, including age or quality. In several passerine species, the production of trills (fast repetitions of notes) positively correlates with age, and thus may serve as a quality signal. In this study, we investigated whether soft trills, a conserved part of the song of the Tree Pipit (Anthus trivialis: Motacillidae), improve in quality (quantified as trill rate and regularity) with age and whether successfully paired males differ in soft trills from single males. We studied two European populations, in Czechia and England, using both a longitudinal approach (changes within the same individuals between consecutive seasons) and a cross-sectional approach (comparison between age categories). There was considerable variation in the rates of soft trills of individual males within the populations studied. Most males only sang regular trills, but irregular trills were recorded in songs of some individuals occasionally or frequently. Returning males mostly increased their soft trill rate with age, but such differences were usually smaller than the variation between individuals. When soft trills of one-year-old and older males were compared, no clear differences in trill rate but better trill consistency in older males were observed. Contrary to our hypothesis, paired and unpaired males did not differ significantly in trill rates. Our study adds to the few that have found ambiguous relationships between structures reflecting song quality and age in passerine birds, and it also points to discrepancies between results from longitudinal and cross-sectional analyses.

P2-69

Active sensing collectives: A multi-frontier study of echolocating groups

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When hundreds and thousands of echolocating bats come together they face a unique sensory and motor challenge. Individuals must track their neighbours, following and avoiding them – while also attempting to make sense of the barrage of echoes and calls arriving from all directions. The fact that bats can even manage to fly and show various kinds of collective behaviour using only sound is a fascinating inverse problem that remains acknowledged but under-investigated due to the logistical and technical challenges involved. Here I will present the vision and results of the newly formed 'Active Sensing Collectives' research group, highlighting the multiple approaches we are taking: swarm robotics, methods development and computational modelling. Swarm robotics provides a controlled testbed to investigate sensorimotor strategies. I will highlight our 'Ro-BAT' platform to investigate passive and active-sensing in swarms. From a methodological perspective studying the dynamics of bat swarms is not trivial, especially its acoustics. I will present our field-deployable multi-sensor rig with its acoustic & thermal cameras, microphone array and LiDAR components. The rig promises to overcome the limitations of previous studies with at most 2 sensors at a time. Data from the rig will also drive the validation of RANSAC-based algorithms to track sources in complex, reverberant audio. Finally, through modelling we have shown that echolocation alone provides minimal but potentially sufficient information to navigate in groups. I will discuss our latest dynamic models which incorporate evolutionary simulations to contextualise the wide variety of observed and theoretically possible sensorimotor behaviours that active-sensing collectives may show.

P2-70

Individual contributions to flock decisions in parrots with fission fusion flock dynamic.

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Flock decision may be based on one individual's information or information communicated from multiple individuals. In species with fission-fusion flock dynamics, such as orange-fronted conures (Eupsittula canicularis), decisions are further challenged, due to the frequent variation in flock composition making potential dominance hierarchies transient. Conures have loud contact calls used for recruiting flocks and soft contact calls used during foraging and prior to leaving a site. We used a two-speaker playback experiment to test the flock decision process. In the first part of the playback, the playback attracted the flock to the area and interacted briefly with the flock to establish contact. This part of the playback mimics the natural interaction between flocks prior to a flock fusion decision. The playback then moved to another speaker, which asks the flock to decide whether to follow the playback or not. We recorded the flock decision process using a microphone array in the area where flocks initially landed. Flocks followed the speaker in about 40 % of the trials. Soft contact calls showed a positive linear relation to flock size, whereas the relation between flock size and call rate indicated that only 1-3 individuals in a flock emitted loud contact calls, reflecting a strong hierarchy on recruitment decisions, whereas decisions on leaving a site appeared collective. Study was funded by Aarhus University Research Fund.

P2-71

Exploring how Northeast Pacific blue whale song characteristics vary with season and location

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Passive acoustic monitoring (PAM) is a powerful tool for assessing baleen whale vocal behavior, providing insights into reproductive activity and migration patterns. Male Northeastern Pacific (NEP) blue whales sing as a reproductive display during Fall and Winter, just before mating season. These songs consist of several phrases, each starting with one A note, then multiple B notes and a short rest. Because B notes are louder, they may require more energy to produce than A notes. Longer and lower frequency B note phrases in a male's song may signal to females the singer is reproductively fit. The goal of our study is to investigate how the energy male NEP blue whales exert during their reproductive display varies seasonally and spatially off the US West Coast. To do this, we will annotate the A and B note timings, B note frequencies, and phrase duration of songs recorded on ~70 ocean bottom seismometers (OBSs) deployed offshore from northern Washington to northern California from 2013 to 2014. OBSs are normally used to record seismic activity but can also record low frequency whale calls. Preliminary results from ten individual songs show a higher number of B notes are included per phrase later in the season and farther south. This is consistent with higher energy reproductive displays occurring just before the reproductive season as whales migrate southward toward their breeding grounds. Characterizing when and where blue whales produce high-energy songs improves insight into their reproductive behavior, helping focus conservation efforts on key reproductive habitats.

P2-72

Social reward function for emotional ultrasonic vocalizations in rat (Rattus norvegicus)

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Rodent ultrasonic vocalizations have a variety of social functions. Of these, rats (Rattus norvegicus) have long been empirically found to emit two types of USVs, with each vocalization following a contrasting emotional context. Short, continuous vocalizations near 50 kHz (50 kHz USV) are produced under the positive emotional state of the individual. Long, steady vocalizations around 22 kHz (22 kHz USV) are produced under unpleasant emotional conditions. I demonstrated that these vocalizations not only reflect the emotional state of the individual, but also transmit emotional information to other individuals who listen to them, resulting in empathic emotional contagion. This experiment was confirmed using a series of protocols called the Cognitive Bias Test. In addition to this, I will measure various social behaviors and brain activities when USVs are played back to rats and attempt to make a comprehensive discussion. As an example, when the approach behavior to the sound source was implemented in the radial maze, the ecological vocalization, USV, caused significantly more approach behavior than the control stimuli. The brain activity during the approach behavior was correlated with the emotional system and the part of the brain controlling social reward including oxytocin system.

P2-73

Development of an acoustic camera system for measuring vocal communication in marmosets

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Common marmosets are an excellent animal model for studying the mechanism of vocal communication due to their rich vocal repertoires and strong social behavior. While short-distance vocal communication is frequently observed, most studies have focused on long-distance vocal communication (i.e., Phee calls), largely due to the difficulty of identifying which individual is vocalizing during close social interaction. Here, we present a novel acoustic camera system – comprising a 16-channel microphone array integrated with a camera – designed to identify the vocalizing individual during social interaction. The system is capable of recording individual vocalizations of freely moving marmosets in a soundproofed chamber. The results show high accuracy in localizing vocalizations and assigning them to specific individuals during short-distance social interaction. Notably, the system revealed that marmosets engage in turn-taking vocal exchanges using trill calls. The peak response latency in these interactions was 500 ms following the offset of the initial call, comparable to dyadic interactions in humans. These findings suggest that marmoset monkeys are a suitable animal model for investigating the evolutionary precursors of human speech. The acoustic camera system provides a promising tool for uncovering the behavioral and neural mechanisms underlying the flexibility of human speech in social contexts.

P2-74

Population density effects calling behavior in grasshoppers (Locusta migratoria)

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Acoustic communication, defined as the interference between sound signals emitted by individuals, play an important role in many groups of insects on the interaction and reproduction since such signals are usually associated with mate attraction and competition. However, the influence of the social structure, such as the number and sex of conspecifics in the same area, on the production of acoustic signals remains poorly understood in many species. The grasshopper Locusta migratoria (Linnaeus, 1758) is an orthoptera species characterized by a phenotypic plasticity that depends on population density. A solitarius and a swarming gregarious phase presenting the two extreme phases which usually involve progressive changes in the behavior of individuals. In our study, we examine how the sex of individuals and social structure influence acoustic signal emission by males and females. Therefore, 24-hour audio recordings are being carried out across six different contexts: 1) single individuals (males or females), 2) female-female pairs, 3) male-male pairs, 4) female-male pairs, 5) threesome (one female and two males), 6) foursome (one female and three males). All the recordings are conducted under controlled laboratory conditions. Our results suggest that single individuals are not calling, and acoustic signals are not used for long-distance communication. However, the number of calls increases in social arrangements, particularly involving multiple males. We expect that the results provide evidence on the communication plasticity in L. migratoria, with potential implications for understanding the mechanisms of interaction between individuals and intra-specific competition.

P2-76

All at sea with seal calling: Long term acoustic tags show that harbor seals communicate far from breeding grounds

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Harbor seals (Phoca vitulina) are vocally versatile and capable of both vocal learning and dialects. During the mating season male harbor seals produce loud and low frequency mating roars that function as either sexual advertisement to attract females or in male-male competition for territory in lek-like displays. Much less is known about male and female vocal behavior outside the mating season with fundamental implications for understanding at-sea social behavior and vulnerability to anthropogenic noise exposure. This study aimed to investigate vocal behavior outside of the mating season to address whether they produce vocalizations to communicate at sea where their communication and active space can be affected by anthropogenic noise. To investigate this, we equipped 7 harbor seals (3 males and 4 females) in the North Sea with animal-borne DTAG4 tags which recorded sound, depth, movement and GPS data. The deployment durations ranged from 7-26 days of continuous recordings and subsets of that data were manually audited aurally and visually to mark vocalizations. We show that both male and female harbor seals vocalize during foraging trips at sea with call rates of less than one vocalization per hour, suggesting fewer and less complex acoustically mediated social interactions at sea. Our findings demonstrate that harbor seal vocalizations in both sexes are not restricted to their mating season and haul-out site, and their frequency ranges overlap with anthropogenic noise. Furthermore, vessel noise induced masking of seal vocalizations and potential decrease in active space should be considered in future impact studies.

P2-77

Individually constrained song pitch and its role as a threat signal in the territorial interactions of African wood doves

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The frequency of acoustic signals typically correlates negatively with body size across animal species. However, within-species comparisons often yield inconsistent or contradictory findings. Our study focuses on two African wood dove species (genus Turtur) living in forest or savannah habitats. Both species exhibit low-frequency songs, with males consistently producing phrases at the same pitch. As their song frequency parameters correlated negatively with some body measurements of males, it may serve as a potential proxy for rival strength. Experimental investigations involved measuring the response of territorial males of both species to computer-manipulated songs having a minimal or maximal frequency within each species' specific range. Responses to simulated unknown stranger intruders were influenced by the frequency of the playback song and by the relative frequency of the focal males compared to the playback frequency used. Notably, males from different habitats exhibited divergent behaviours. Blue-headed wood-doves from Ghanaian lowland rainforests tended to avoid approaching low-frequency song playbacks, presumably indicating the presence of a large intruder. In contrast, Emerald-spotted wood doves from Ugandan savannahs displayed responses that were more reliant on the disparity between a stranger's song frequency and their own. These findings are examined concerning the mechanisms that underlie song production and the differences in song propagation and territorial defence strategies between forest and savannah habitats.

P2-78

Machine learning to separate individual budgerigar vocalizations in a social setting

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Using artificial intelligence to analyze audiovisual footage of animal groups holds great potential for studying social behaviors where manual analysis of recordings is impractical or even impossible. Our work combines computer vision, machine learning for audio, and acoustic methods to automatically separate individual vocalizations from recordings of a group of animals. In an aviary housing 11 budgerigars (Melopsi(acus undulatus), the approach starts with a computer vision system designed to localise and identify individual birds on video. This system was developed by recording each bird individually to generate training data with minimal manual effort and by fine-tuning YOLOv11 and MobileNetV2 models, enabling the detection and identification of budgerigars with 98% mAP and up to 94% accuracy, respectively. At the same time, audio recordings of the group's vocalization are taken with a planar array comprising 112 microphones. The spatial position of individual birds, derived from video, is integrated with acoustic source localization techniques performed on the audio to separate vocalizations from recordings where up to 11 birds vocalize simultaneously, and to attribute them to specific individuals. This pipeline allows us to capture individual vocalizations difficult to record outside of a group, such as a female budgerigar's. Additionally, it will be used to study turn-taking behaviour and consonant-, and vowel-like elements in budgerigars. Importantly, such an automated system is adaptable across species and enables the creation of large datasets of individually labelled social vocalizations, which allows for deeper analysis of non-human vocal interactions in naturalistic seangs on both a structural and behavioral level.

P2-79

Social feedback beyond tutors: Insights from a vocal learning bat

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The Greater sac-winged bat's vocal development is a social learning process that includes a conspicuous vocal practice behaviour, known as pup babbling, lasting for 7-8 weeks. This babbling behaviour involves the learning and rehearsing of song syllables acquired through vocal imitation of adult male tutors. Babbling is expressed in long multisyllabic vocal sequences arranged in so-called babbling bouts, with durations of up to 43 minutes. Maternal behaviours can accompany these babbling bouts. A detailed investigation of the vocal ontogeny of 19 wild pups in Panama and Costa Rica revealed that maternal feedback significantly influences the duration of vocal practice and the presence, versatility and maturity of learned song syllables. This is the first study demonstrating that non-tutorial social feedback significantly influences song learning and vocal practice in a non-human mammal. The findings underscore the importance of studying vocal learning in natural environments to better understand how social dynamics shape learning processes. This research provides a foundation for more in-depth investigations into vocal development in the Greater sac-winged bat and offers new possibilities for comparative studies across mammalian vocal learners.

P2-80

Learning without a model: Non-imitative vocal production learning in a male grey seal

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Vocal production learning is a rare skill that requires elaborate control over the sound production system. This control is often either laryngeal or supra-laryngeal; in a few species, such humans, it involves both. Grey seals (Halichoerus grypus) exert supra-laryngeal control in sound production. Can they also control their vocalizations via the larynx? We trained an adult seal to actively modify fundamental frequency (f0) properties of his vocalization. Through a semi-automatic operant conditioning paradigm, we test whether the seal could significantly increase the vocalization's delta f0, i.e., the difference in f0 between the vocalization's end and its start. The results demonstrate strong evidence of phonatory production learning, as the seal successfully adjusted several f0 measurements to meet the increasing delta f0 requirements. Specifically, the seal increased his vocalization's delta f0 from a pre-experimental median of 22.6 Hz to a median of 345 Hz in the final session. Further, our data show the seal used a non-obvious strategy to solve the task, where he increased the vocalizations' delta f0 by lowering the f0 at the start of the vocalization and raising the f0 throughout the vocalization. The slope and duration of the vocalizations both increased throughout the experiment, showing that achieving higher delta f0 values was not the result of adjusting a single parameter. Our findings imply that grey seals have considerable control over their larynx, which complements previous evidence of supra-laryngeal control in this species.

P2-81

A meta-analysis on the developmental stress hypothesis in songbirds

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Is song learning ability an honest indicator of individuals' condition? To answer this question, the developmental stress hypothesis was proposed. It predicts that exposure to stressors during the developmental period impairs vocal learning in songbirds. Testing this idea, many previous studies investigated the effects of a variety of stressors on song traits or nuclei, which included food restriction, corticosterone administration, or sibling competition, but reported mixed outcomes. Therefore, we conducted a meta-analysis that included over 30 papers examining the effects of various stressors on birdsong development across more than 10 species. Overall, we could not find strong evidence that the stressors affected birdsong. The aggregated effect size was small, not statistically significant, and showed a sign of publication bias. Interestingly, however, nutritional conditions had a slightly greater impact on songs than other stress factors, suggesting the importance of nutritional stress on song learning. As there were limitations on the number of primary studies and species, the conclusion of the analysis may be restricted. We are still collecting publications on this topic and will report updated results in the presentation. We will also discuss the contrast between the studies examining the natural variations and those with experimental manipulations of stress factors.

P2-82

Do nestling songbirds prefer simple songs?

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Song learning in songbirds is a critical process for sexual selection and cultural evolution, with early sensory learning guiding later song production and preferences. In pied flycatcher (Ficedula hypoleuca), males produce long, complex songs before pairing and shift to simpler, shorter songs during the nestling stage, potentially providing early auditory input for nestlings. While nestlings have been shown to discriminate conspecific songs, local dialects, and even their social fathers' complex songs at an early age, their responses to these familiar simplified songs versus unfamiliar complex songs remain unknown. Nestlings may exhibit a stronger response to shorter songs either because these are the songs they commonly hear or because they are inherently easier to process and recognize. We conducted playback experiments presenting nestlings with both simplified and complex songs from the same male. Our results revealed that nestlings responded more actively, with increased postural begging behaviors, to shorter songs than to longer songs, a preference that persisted regardless of prior exposure. These findings suggest that both early auditory experience and inherent acoustic salience contribute to song recognition and learning, highlighting the role of early song exposure in shaping vocal development.

P2-83

Good song learners die young and poor learners survive long

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One of the unresolved questions regarding the evolution of birdsong is how much cost vocal learning imposes on individuals. Is learning costly?–This question is crucial when we attempt to understand whether song-learning ability evolved via the handicap principle or Fisherian runaway. Most previous studies have tested the former by investigating the condition-dependence (honest signaling) of songs. Especially developmental stress on the quality of songs is well determined. However, the association between song learning and long-term fitness remains obscure. Therefore, we focused on longevity in the present study. We have been keeping records of the pedigree and life history of the lab-bred population of Java sparrows (Padda oryzivora) that died naturally and archived their songs for more than 15 years. Using these data, we examined whether better song learners lived longer. Java sparrow males learn songs during development, where social interactions play a vital role, meaning that they learn songs from fathers but not from other conspecifics nearby. We determined the songlearning performance by comparing songs between tutors (social fathers) and tutees (sons) and conducted survival analyses to see if good learners were long-lived. Surprisingly, we found that poor song learners showed better survival (long-lived) while good learners died early. This result is contrary to what is expected from the conditiondependence of songs and supports the idea of Fisherian evolution. We will discuss its possible interpretations and future implications in the presentation.